
AE4135 – ROTOR/WAKE AERODYNAMICS

Airfoil aerodynamics

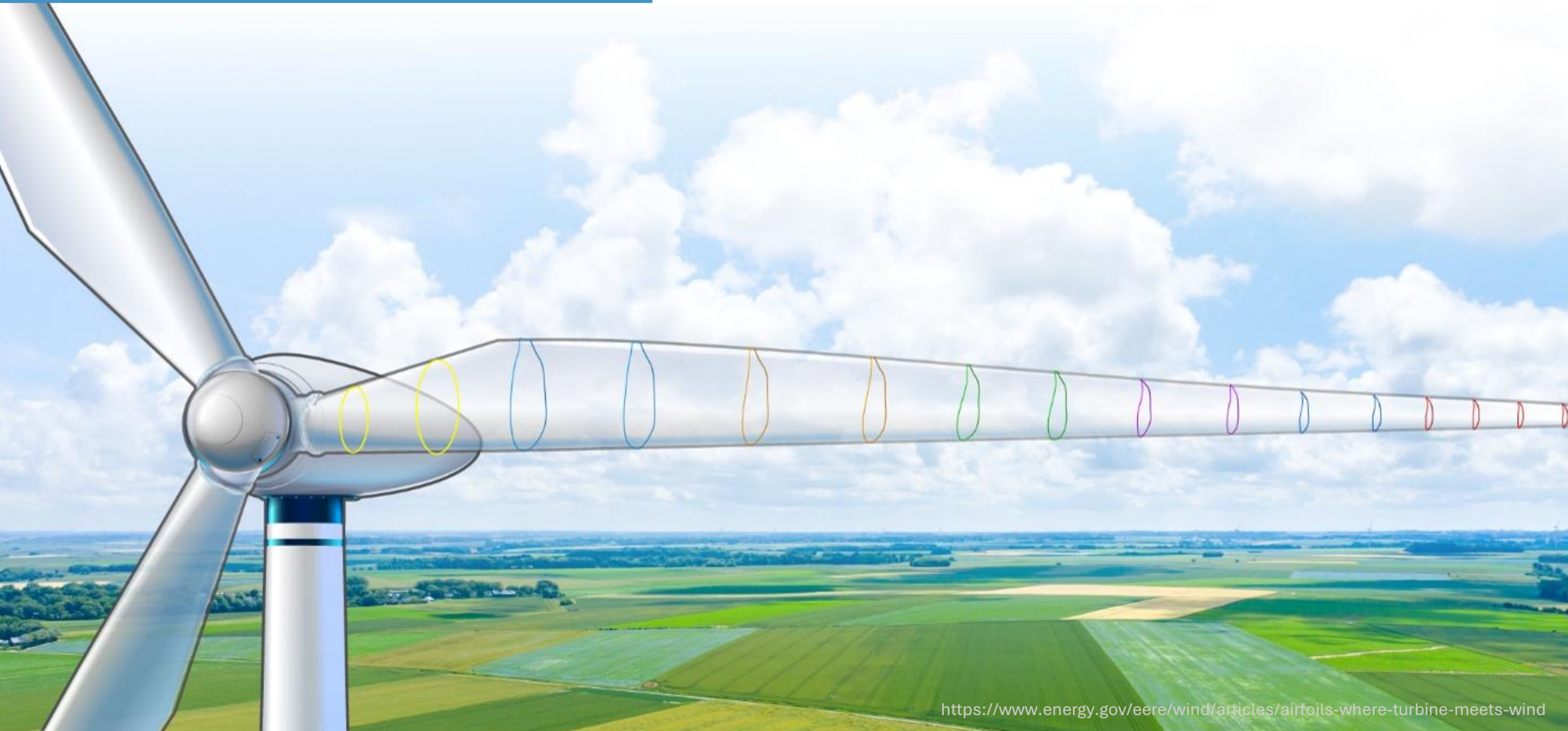
LECTURER



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Assistant professor
Large rotor flow physics

AIRFOIL AERODYNAMICS



LEARNING OBJECTIVES

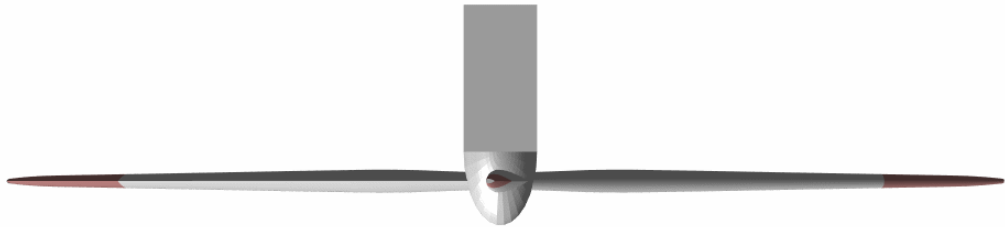
At the end of this lecture, you should be able to:

- Describe the 2D velocity and force triangle
- Derive the desired airfoil characteristics along the blade
- Understand the role of inflow properties on airfoil behaviour
- Understand the role of roughness on airfoil performance
- Describe the approach for airfoil design and evaluation

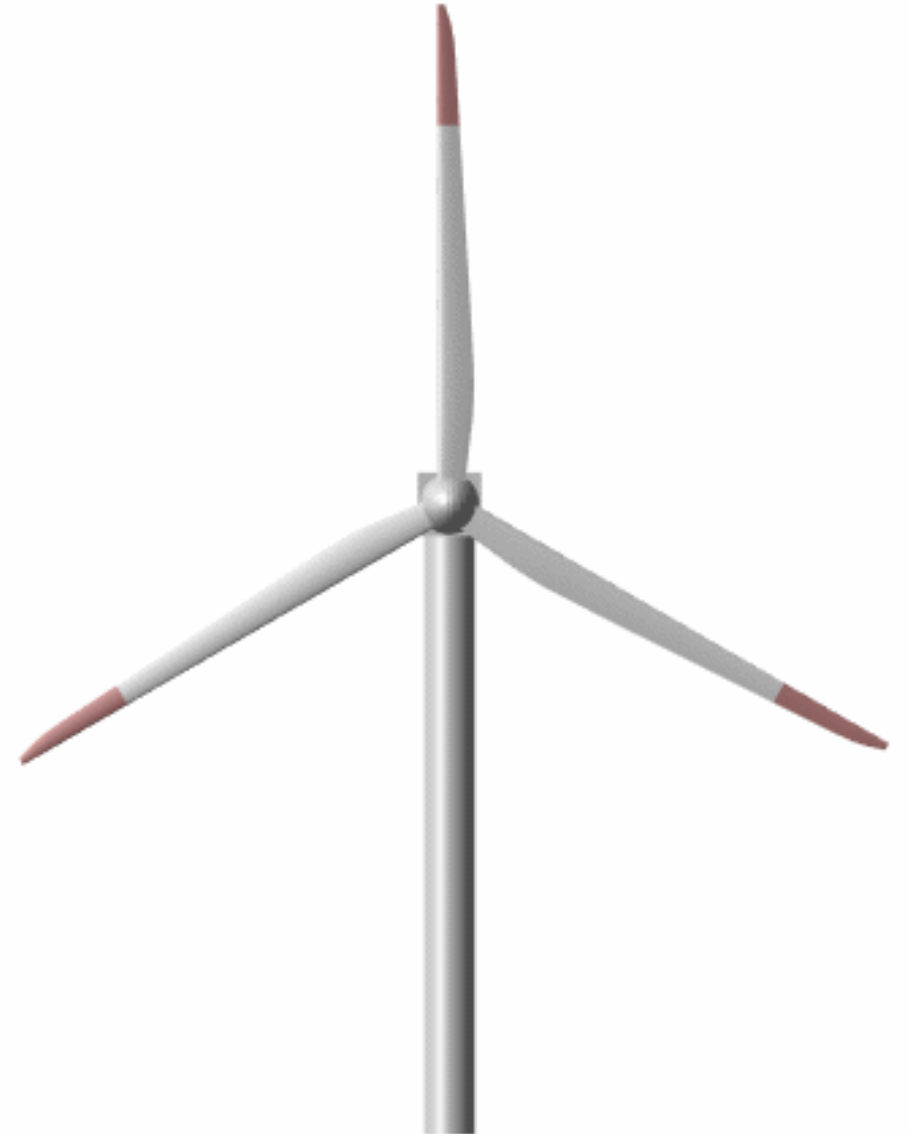
Keywords: (1) Velocity and force triangle, (2) Airfoil requirements, (3) Flow properties, (4) Roughness, (5) Airfoil design and evaluation

VELOCITY AND FORCE TRIANGLE

WIND TURBINE

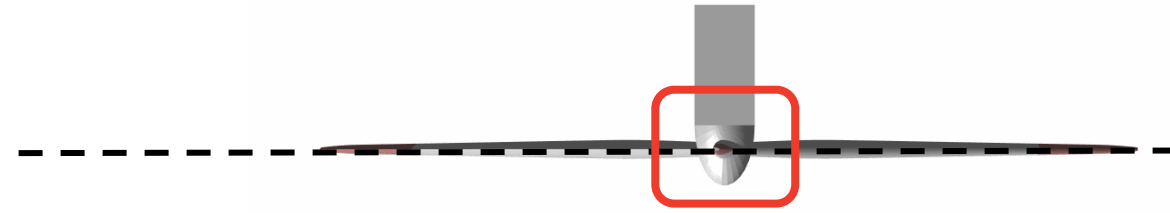


Top view



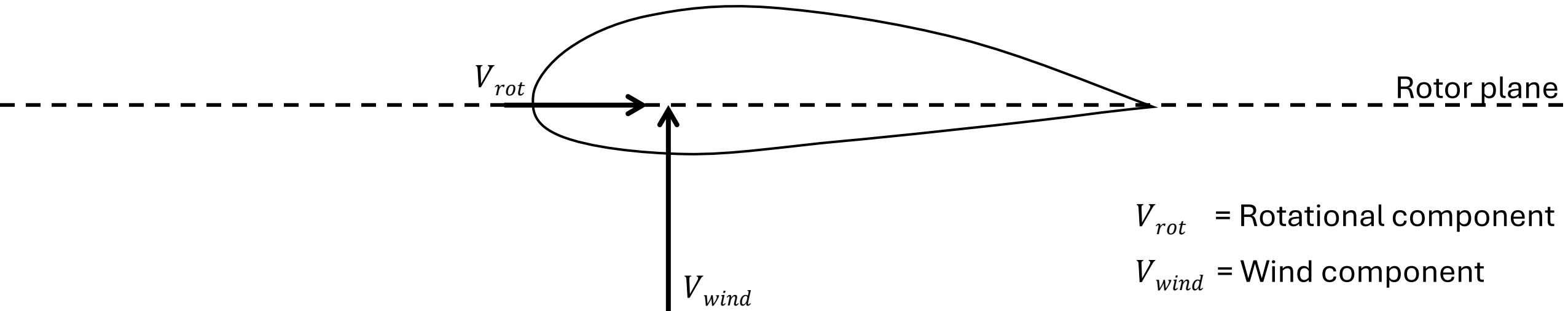
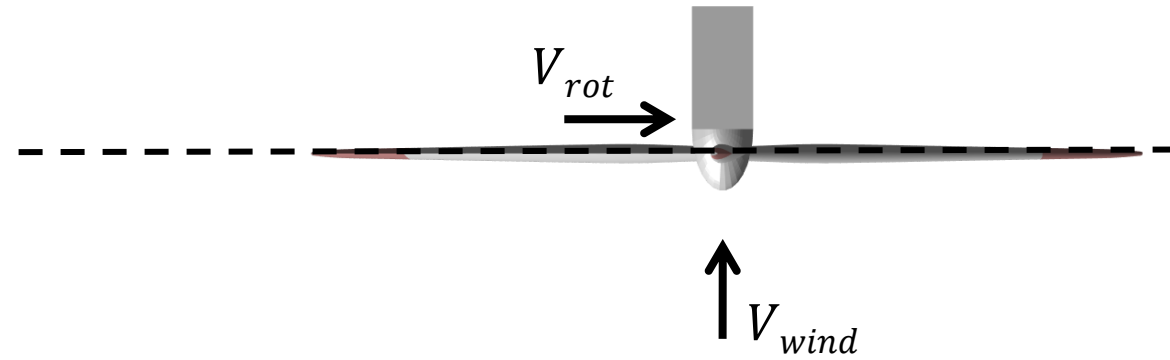
Front view

WIND TRIANGLE

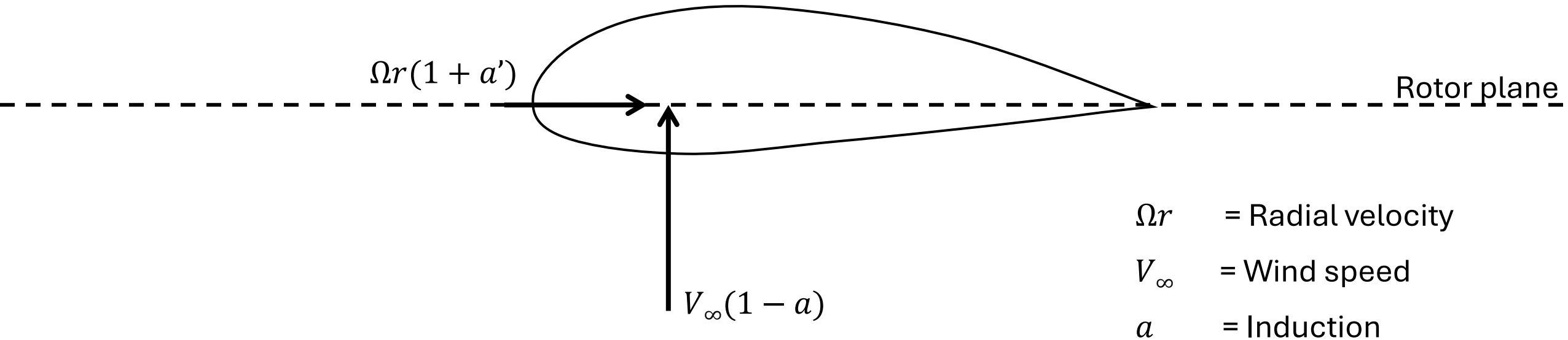
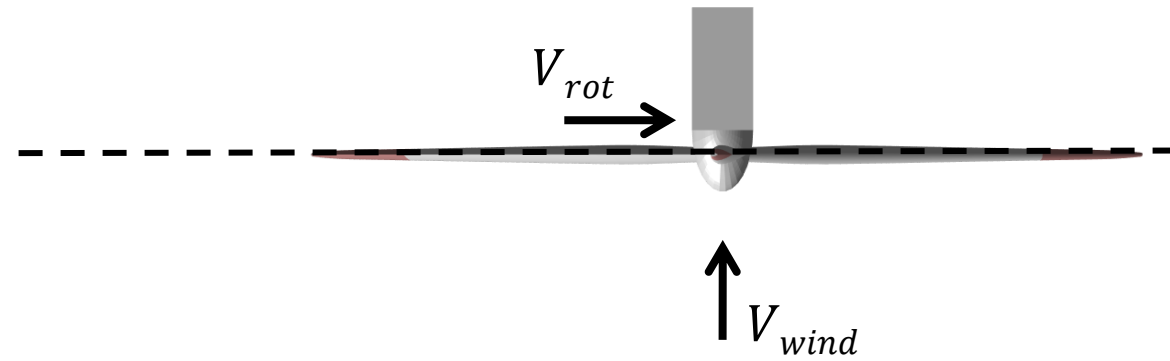


Rotor plane

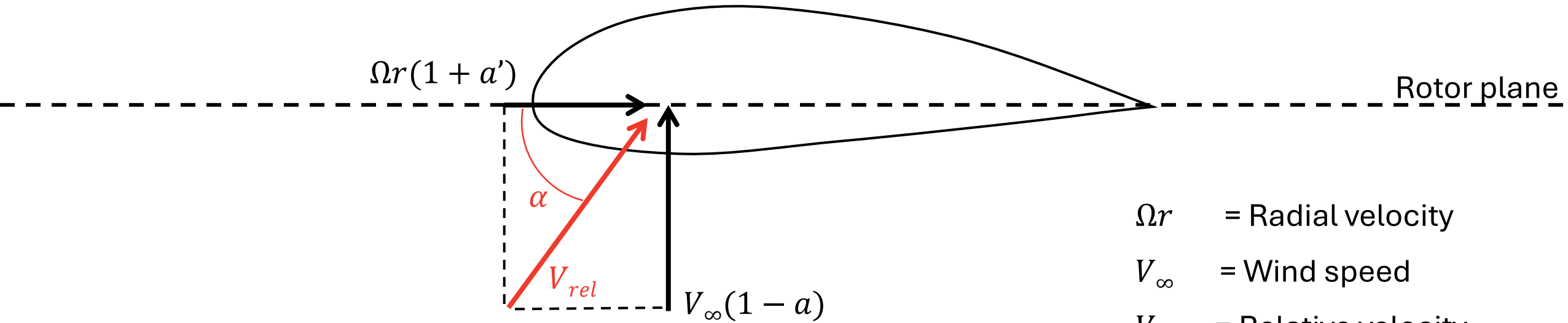
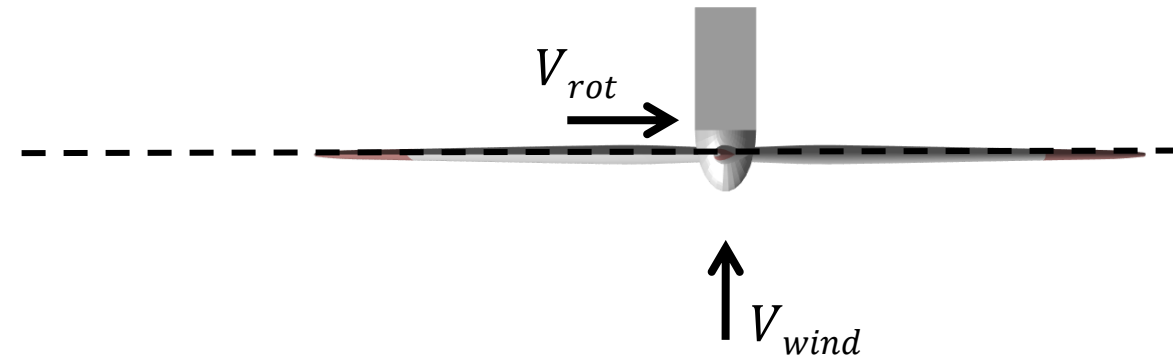
WIND TRIANGLE



WIND TRIANGLE

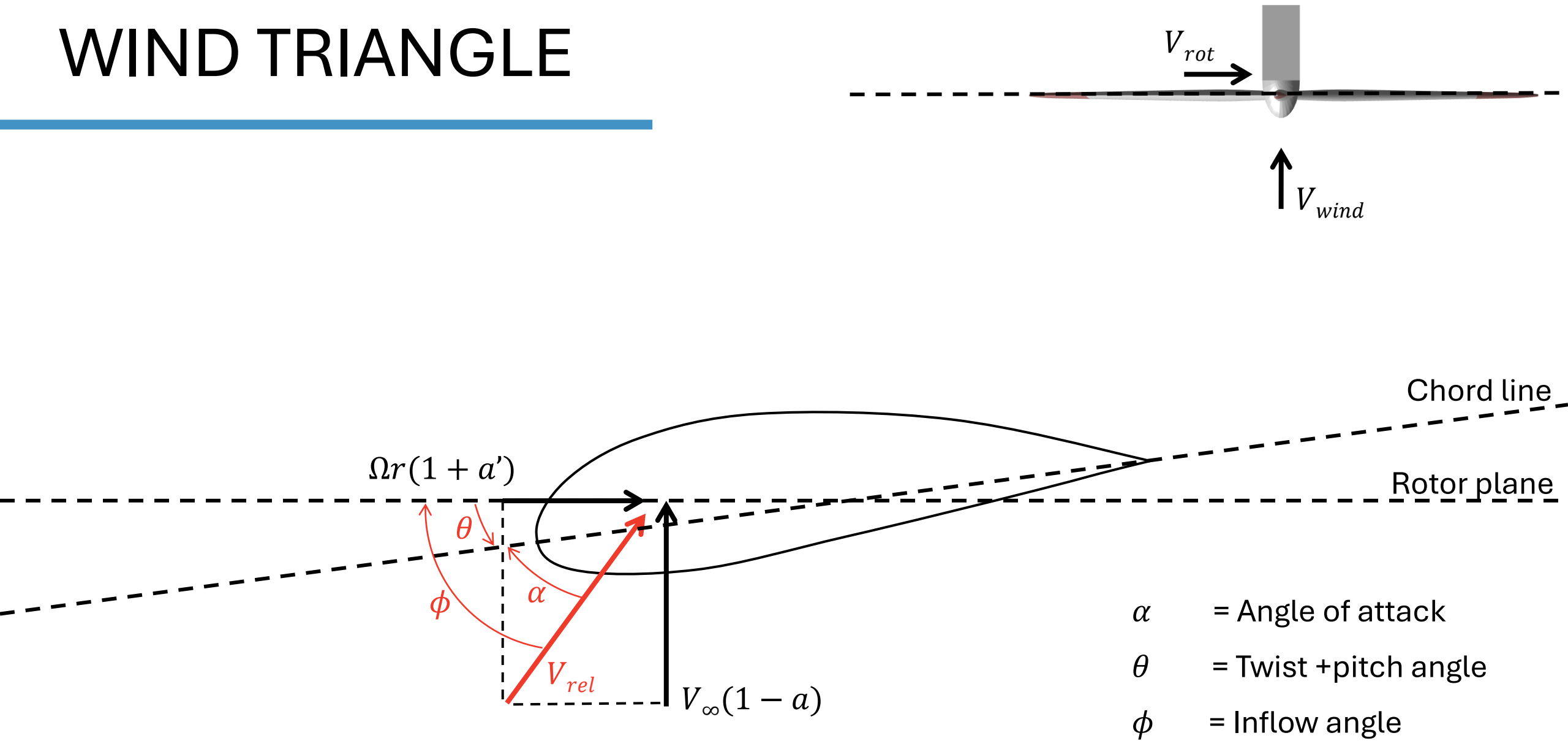


WIND TRIANGLE

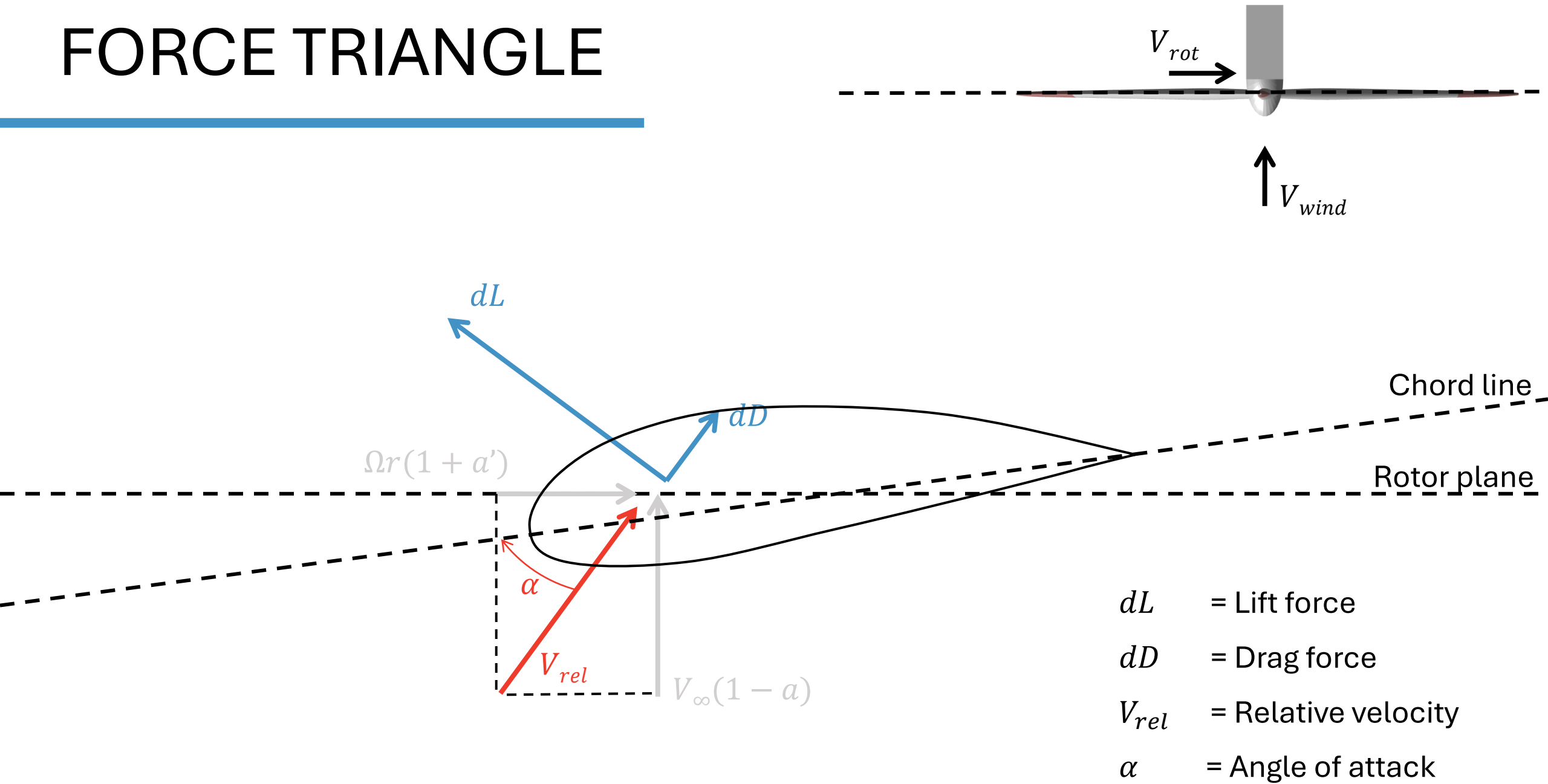


- Ωr = Radial velocity
- V_{∞} = Wind speed
- V_{rel} = Relative velocity
- α = Angle of attack

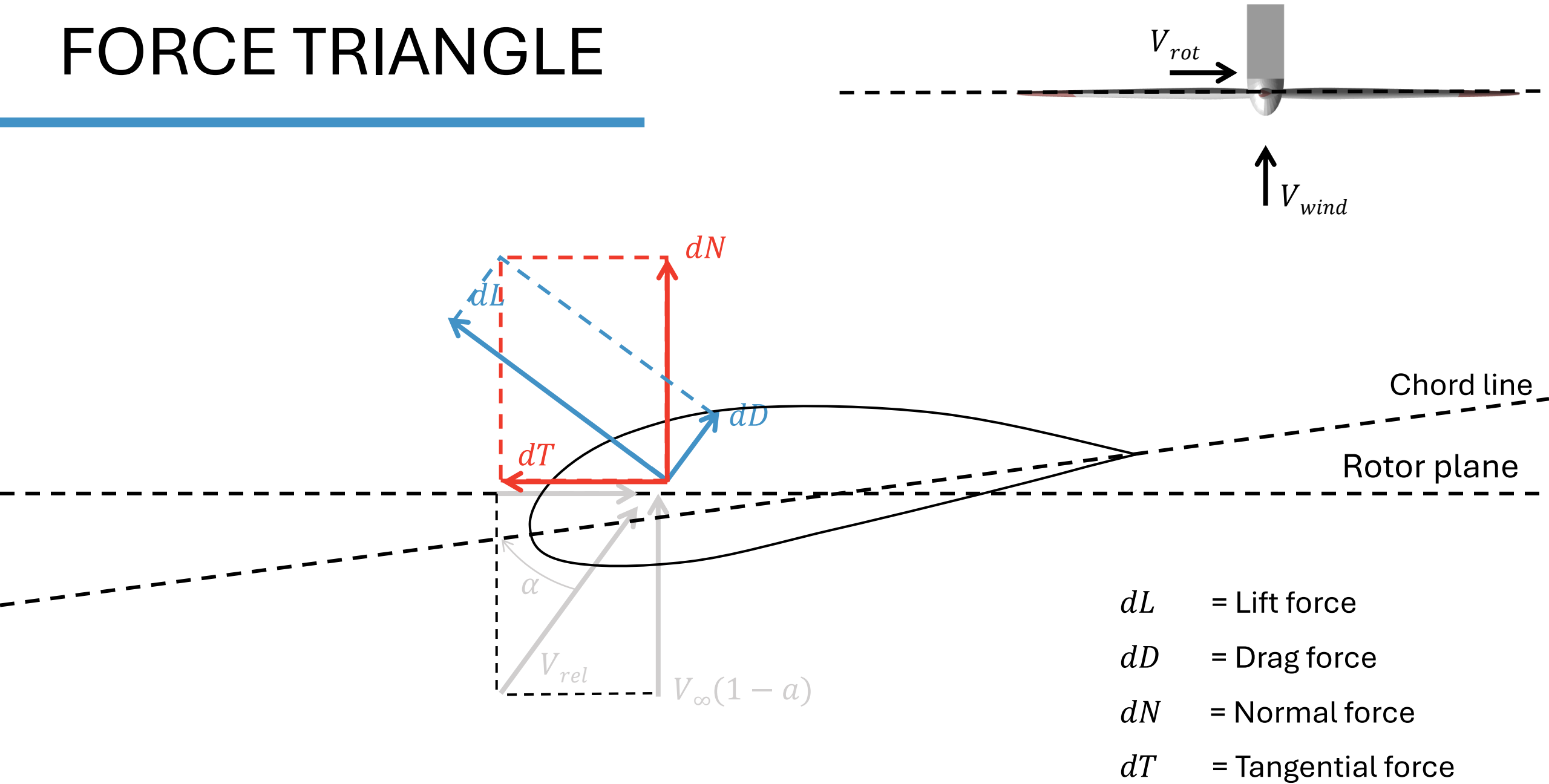
WIND TRIANGLE



FORCE TRIANGLE



FORCE TRIANGLE



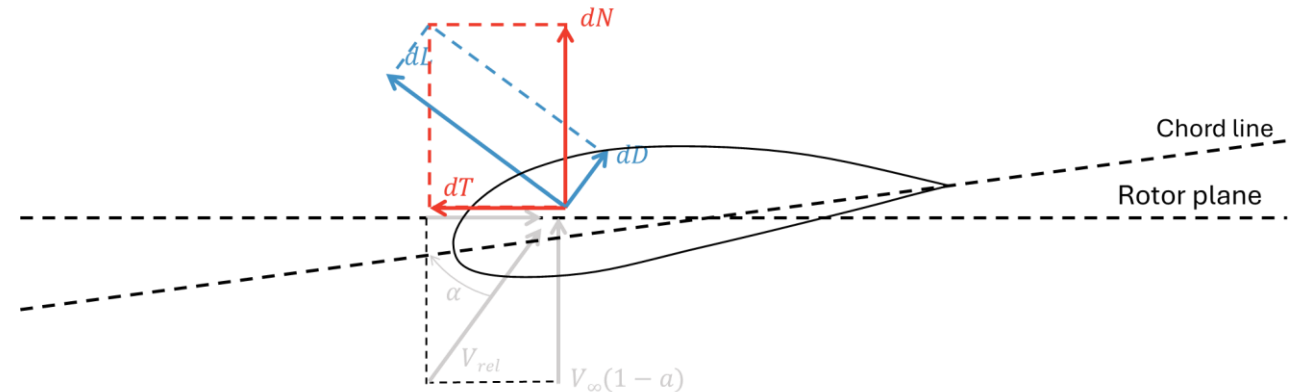
FORCE TRIANGLE

The force vectors on a small spanwise element dr of the blade is:

$$dL = C_l \frac{1}{2} \rho V_{rel}^2 c dr$$

$$dD = C_d \frac{1}{2} \rho V_{rel}^2 c dr$$

c is the local chord on a radial distance r from the rotor axis.



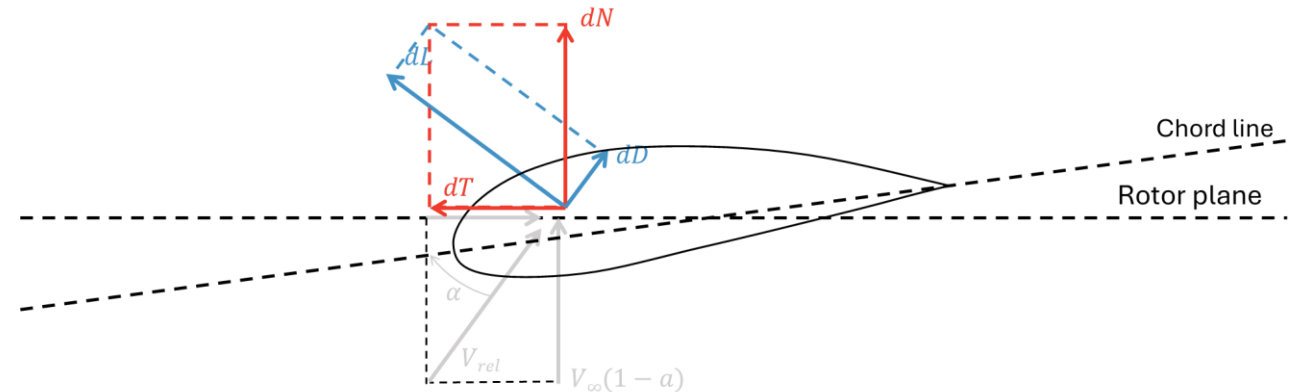
FORCE TRIANGLE

If we decompose the resultant force dR in a component in the rotor plane (dT) and one perpendicular to the rotor plane (dN), we find:

$$dT = dL \sin(\phi) - dD \cos(\phi)$$

$$dN = dL \cos(\phi) + dD \sin(\phi)$$

Here ϕ is the inflow angle (sum of angle of attack and twist/pitch angle)



FORCE TRIANGLE

Combining both set of equations

$$dL = C_l \frac{1}{2} \rho V_{rel}^2 c dr$$

$$dD = C_d \frac{1}{2} \rho V_{rel}^2 c dr$$

$$dT = dL \sin(\phi) - dD \cos(\phi)$$

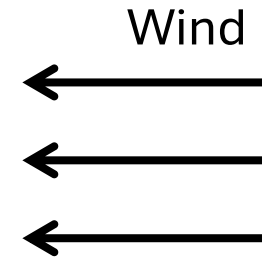
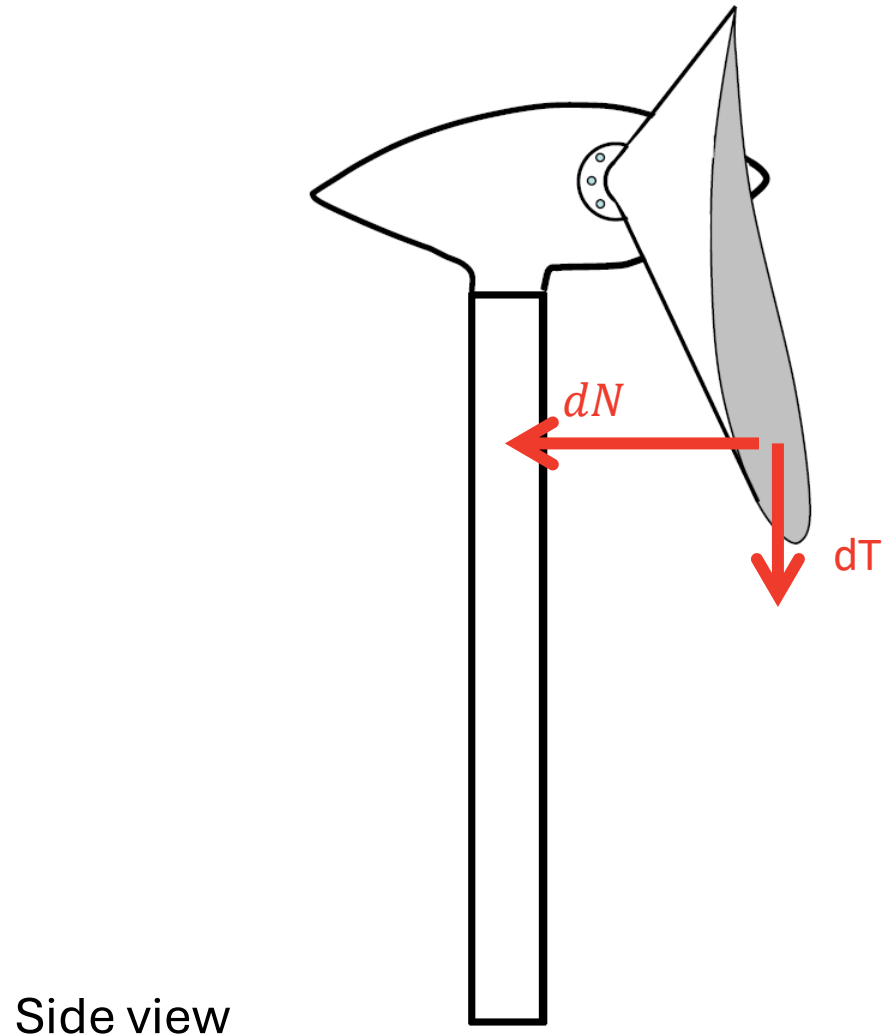
$$dN = dL \cos(\phi) + dD \sin(\phi)$$

Gives the following expression:

$$dT = \frac{1}{2} \rho V_{rel}^2 C_l c \left(\sin(\phi) - \frac{\cos(\phi)}{C_l/C_d} \right) dr \quad \longrightarrow \quad \text{Rotor torque}$$
$$dN = \frac{1}{2} \rho V_{rel}^2 C_l c \left(\cos(\phi) - \frac{\sin(\phi)}{C_l/C_d} \right) dr \quad \longrightarrow \quad \text{Bending force}$$

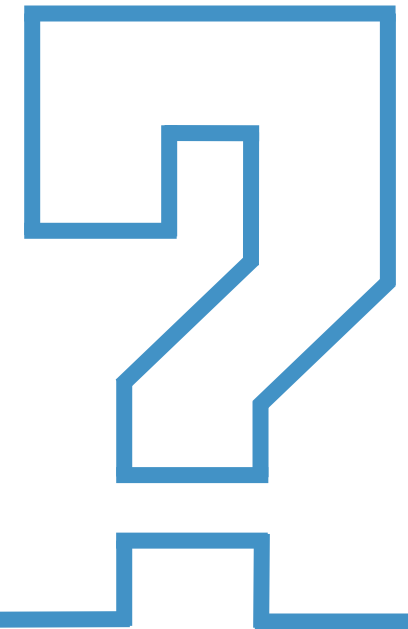
Try this at home!

WIND TURBINE



$$dT = \frac{1}{2} \rho V_{rel}^2 C_l c \left(\sin(\phi) - \frac{\cos(\phi)}{C_l/C_d} \right) dr$$
$$dN = \frac{1}{2} \rho V_{rel}^2 C_l c \left(\cos(\phi) - \frac{\sin(\phi)}{C_l/C_d} \right) dr$$

ANY QUESTIONS ON VELOCITY
AND FORCE TRIANGLE?



AIRFOIL REQUIREMENTS

AIRFOIL REQUIREMENTS (in short)

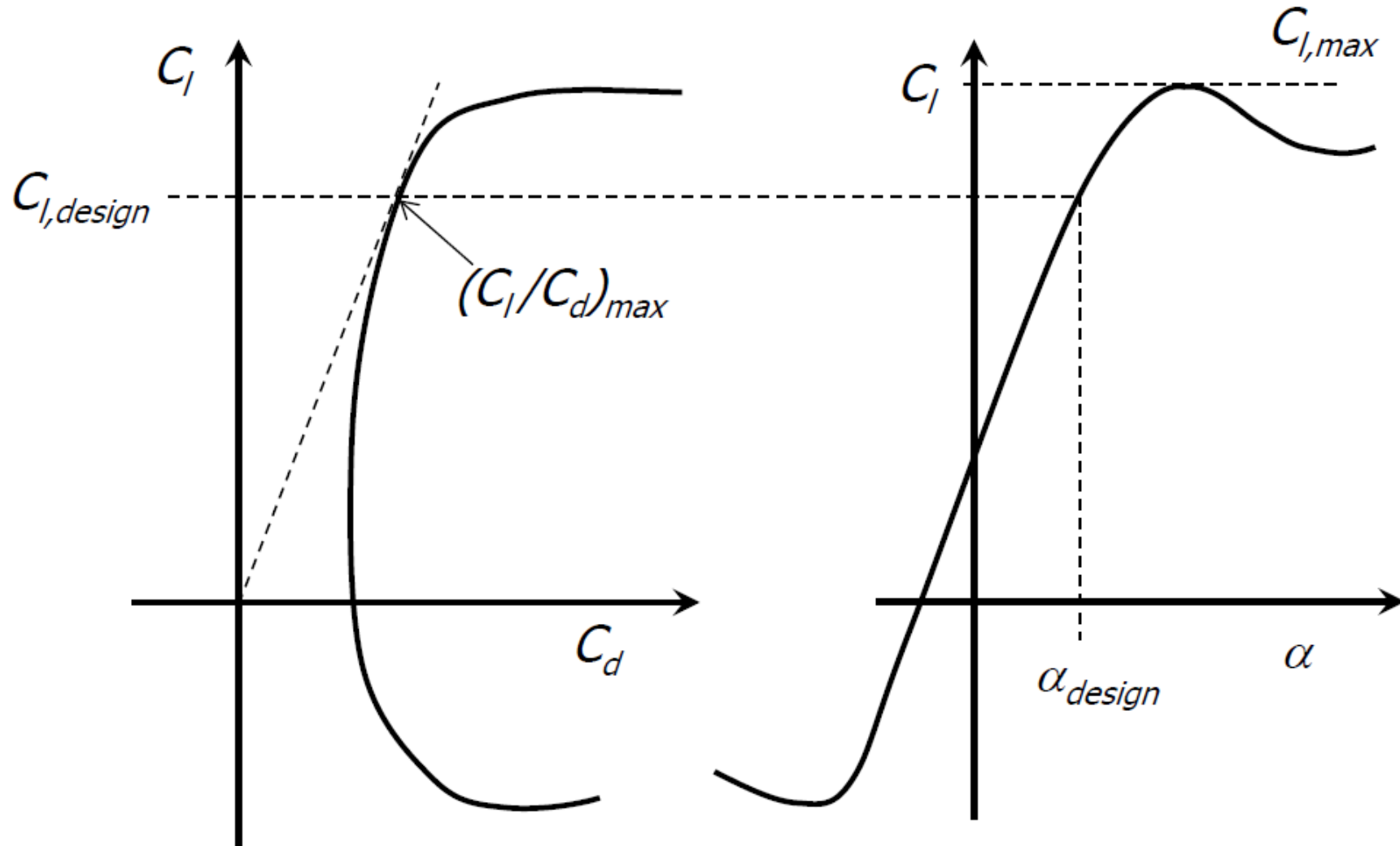
Both torque and bending forces are governed by the term $C_l c$ and the lift to drag ratio C_l/C_d :

$$\text{Rotor torque} \quad dT = \frac{1}{2} \rho V_{rel}^2 C_l c \left(\sin(\phi) - \frac{\cos(\phi)}{C_l/C_d} \right) dr$$

$$\text{Bending force} \quad dN = \frac{1}{2} \rho V_{rel}^2 C_l c \left(\cos(\phi) - \frac{\sin(\phi)}{C_l/C_d} \right) dr$$

- To maximize the torque and minimize the bending moment, a high C_l/C_d ratio of the airfoil is desirable.
- The term $C_l c$ needs to be at a certain level to obtain an optimum deceleration of the wind speed. Maximum power will be obtained when the flow is decelerated by approximately 1/3 (momentum theory).

AIRFOIL REQUIREMENTS (in short)



AIRFOIL REQUIREMENTS (in long)

Considering only maximizing torque or power, we want:

- High C_l/C_d

But there is more:

- Design angle of attack reasonably below $C_{l,max}$
- Thick airfoils give better structural properties (and less weight)
- Insensitive to roughness
- Smooth stall properties (at least for pitch-regulated turbines)
- Low noise: thin trailing edge and boundary layer thickness
- Manufacturing limitations
- Etc.

→ Balancing exercise,
dependent on airfoil
location along the blade

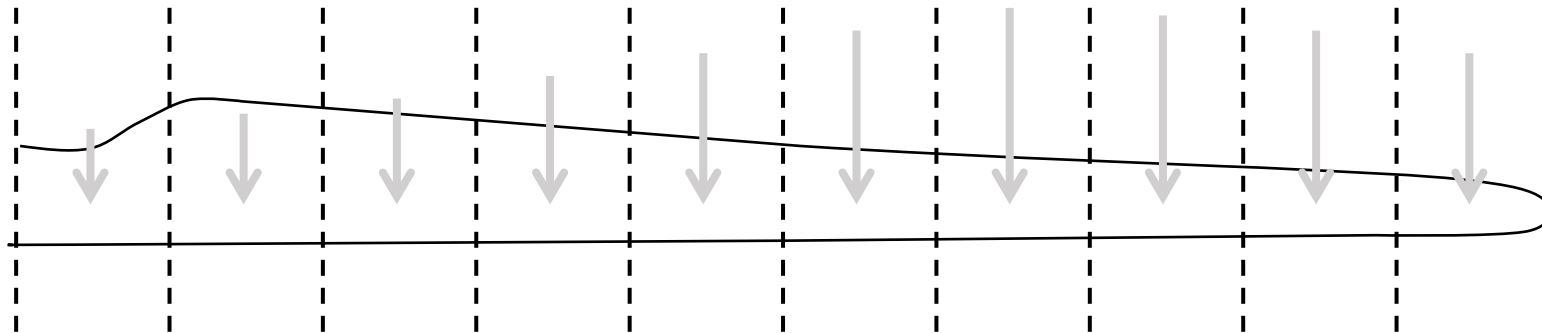
AIRFOIL REQUIREMENTS – GENERAL

For all sections, it holds:

- High C_l/C_d below $C_{l,max}$
- Roughness insensitivity
 - No large reduction in $C_{l,design}$ and C_l/C_d
 - Loss in power is proportional to loss in C_l

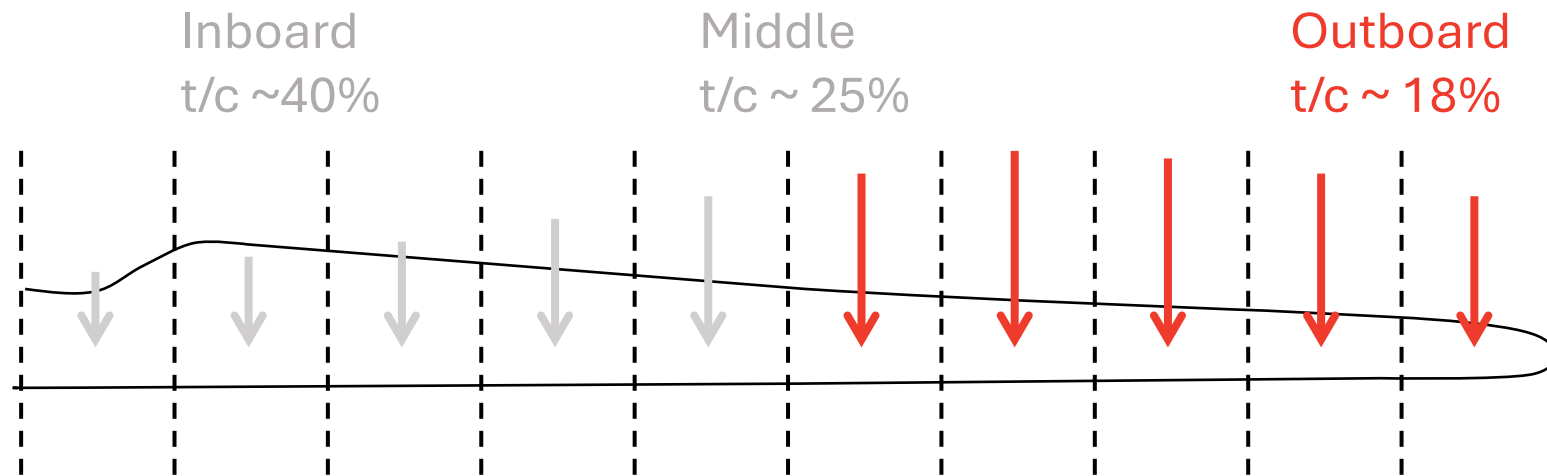
Most designs converge towards:

- All airfoils with $18\% \leq t/c \leq 40\%$
- $C_{l,design}$ between ~ 1.1 and 1.2
- $C_{l,max} - C_{l,design} \approx 0.4$ or about 5 degrees for the associated angles



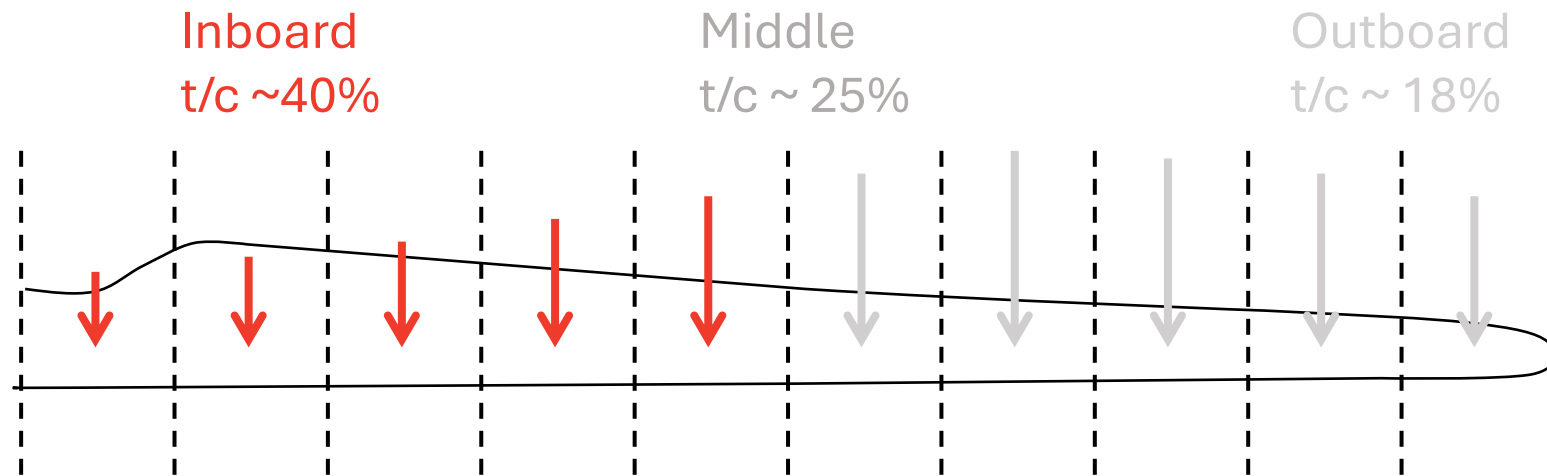
AIRFOIL REQUIREMENTS – OUTBOARD

- Design angle of attack reasonably below $C_{l,max}$ (pitch-regulated turbine!)
 - Far enough to stay away from stall in case of turbulent inflow
 - Not too far/high to limit excessive loads in case of gusts
- Smooth stall behavior (for stall regulated turbines!)
 - To reduce fatigue loads
- No thick trailing edges
 - To reduce noise



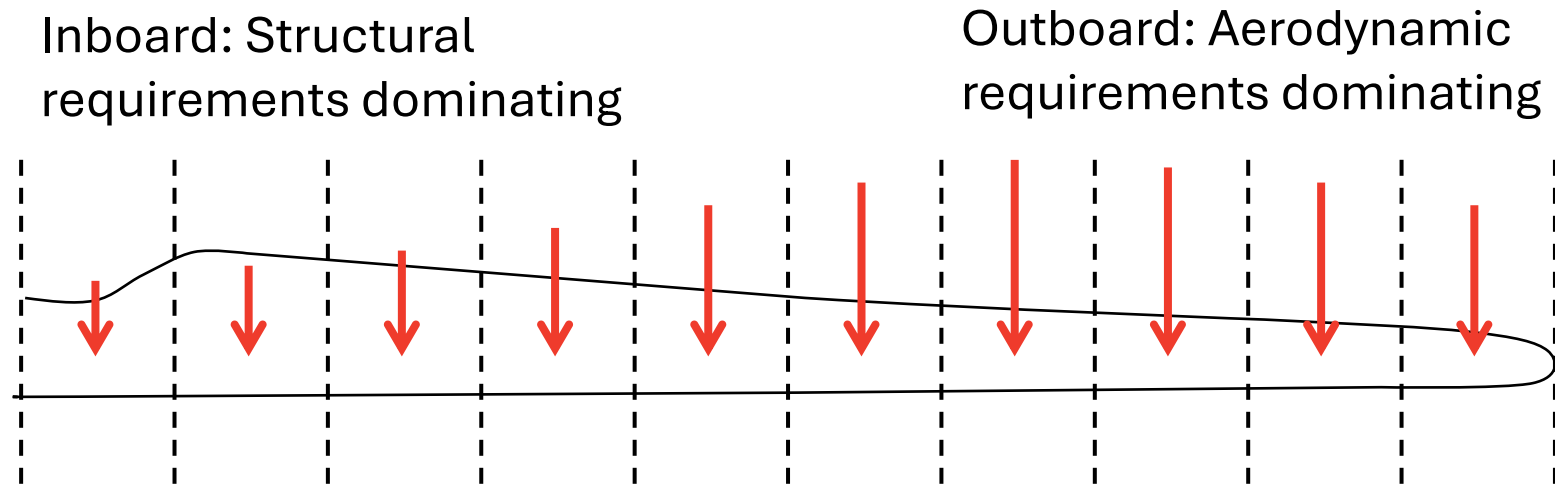
AIRFOIL REQUIREMENTS – INBOARD

- High relative thickness is required
To keep down structural loads and weight
- High $C_{l,design}$
Small root chords and thus avoid problems for parked loads and during transportation
- Benign stall characteristics to reduce fatigue loads



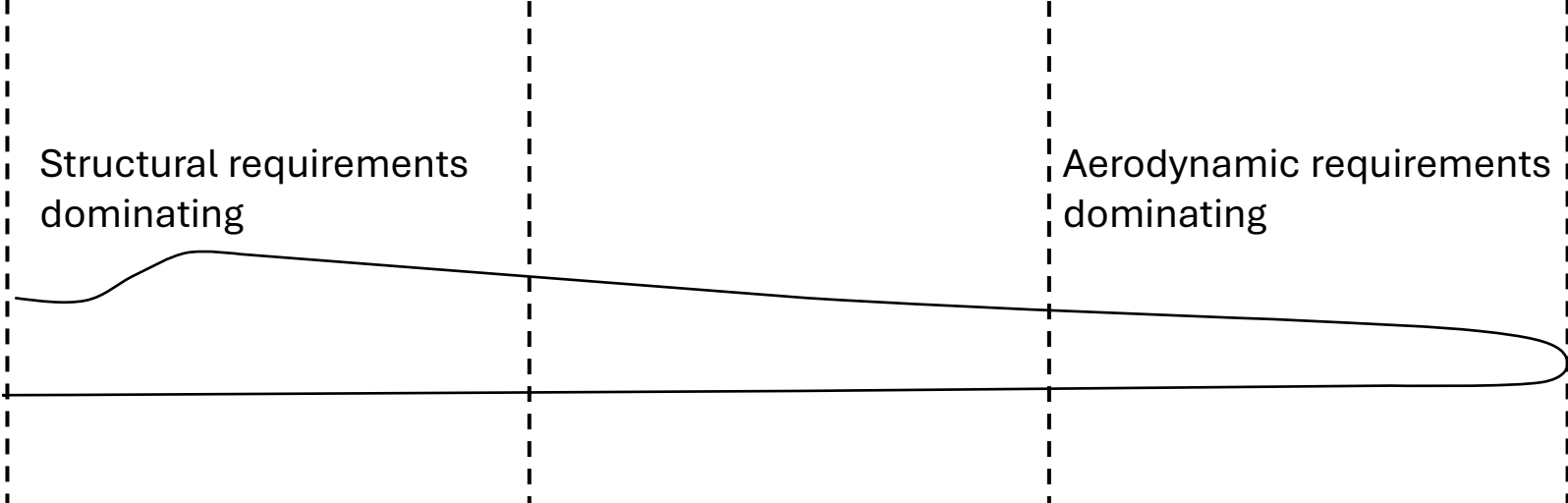
AIRFOIL REQUIREMENTS – BALANCING

- High C_l/C_d vs large thickness
- High $C_{l,design}$ vs smooth stall characteristics

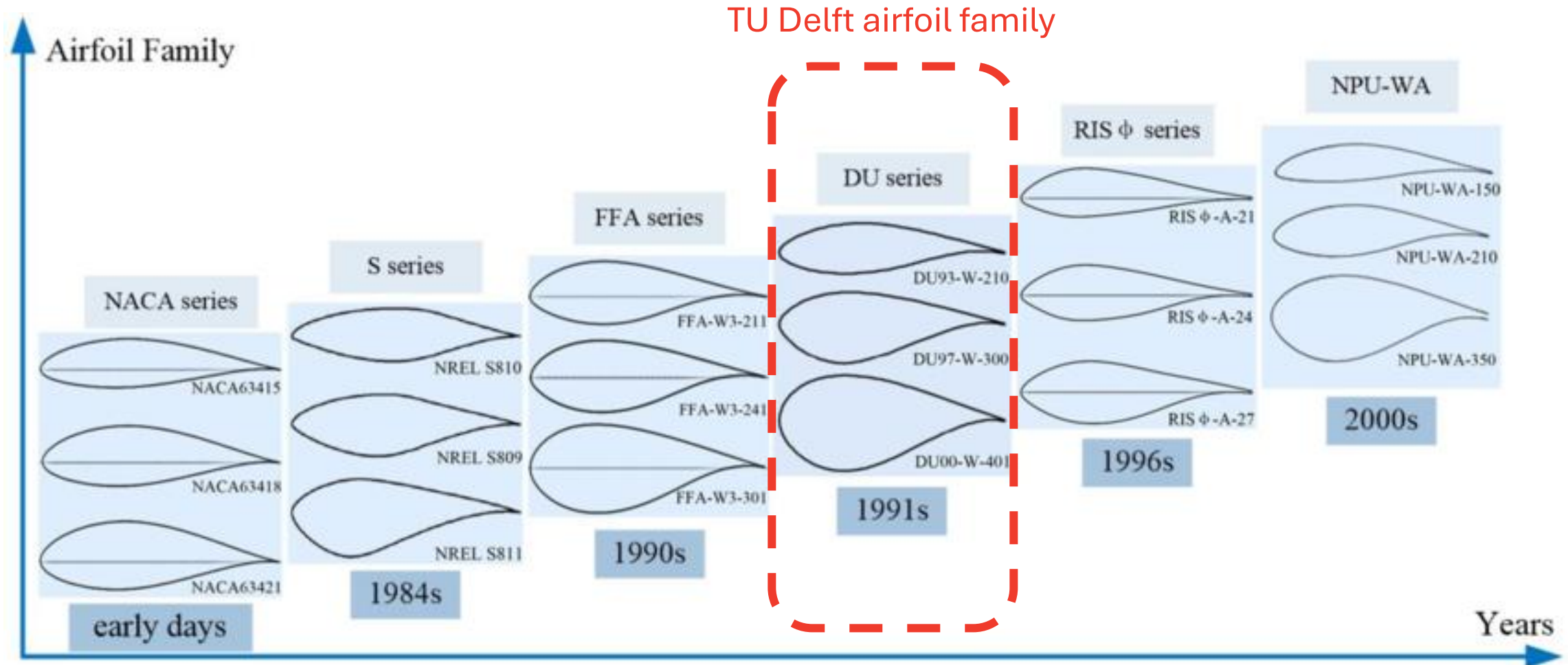


AIRFOIL REQUIREMENTS – BALANCING

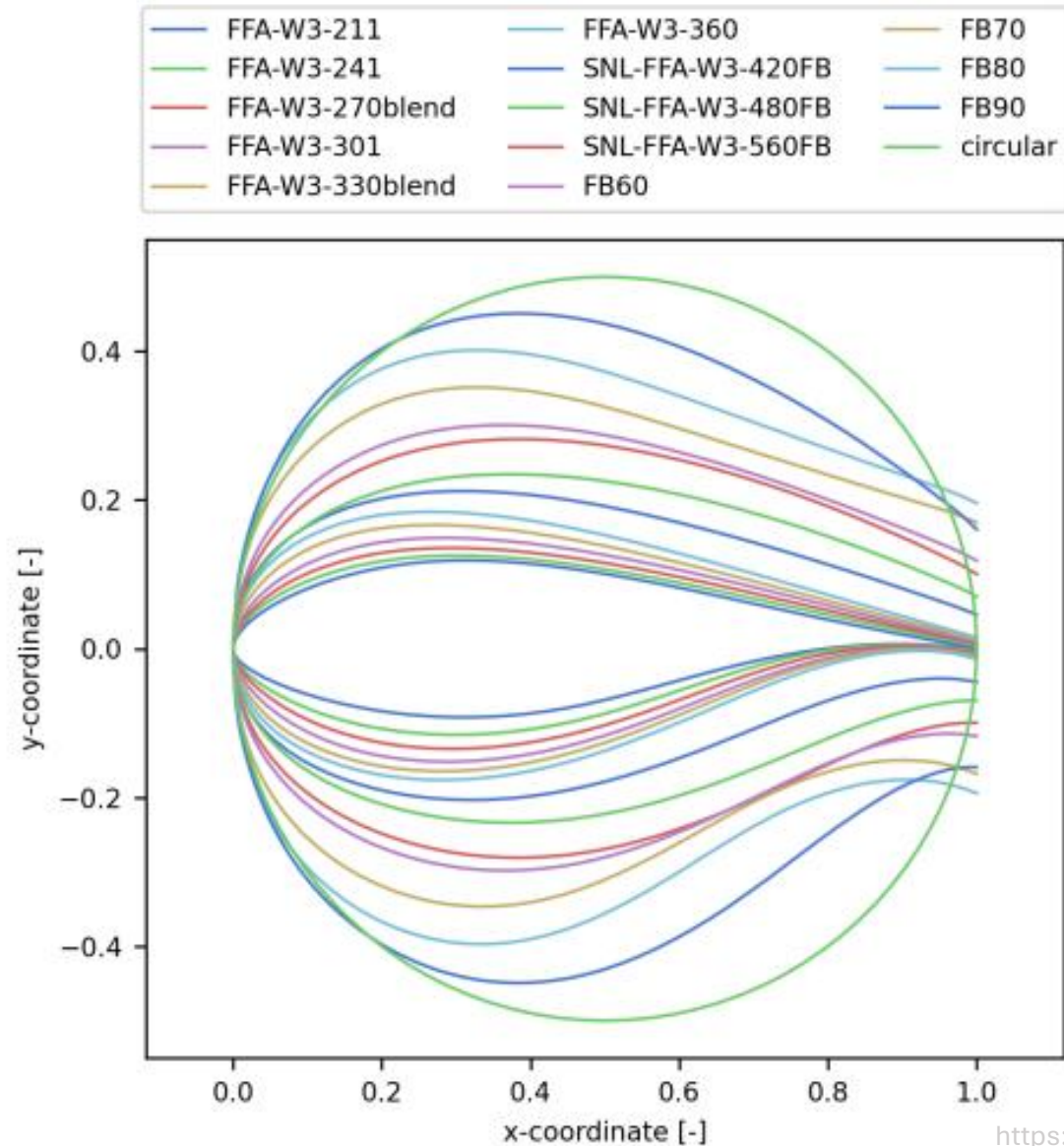
Thickness-to-chord ratio	$t/c > 28\%$	$21\% < t/c < 28\%$	$t/c < 21\%$
High C_l/C_d	●	● ●	● ● ●
Low $C_{l,max}$ and smooth stall			● ●
Insensitive to roughness	●	● ●	● ● ●
Low noise		●	● ● ●
Geometry compatibility	● ●	● ●	● ●
Structural demands	● ● ●	● ●	●
Structural requirements dominating Aerodynamic requirements dominating			

A diagram of an airfoil cross-section is shown at the bottom of the table. It is divided into three vertical sections by dashed lines, corresponding to the thickness-to-chord ratio ranges in the table above. The airfoil has a thick leading edge on the left, a moderate thickness in the middle, and tapers to a thin trailing edge on the right.

AIRFOIL DESIGN



AIRFOIL DESIGN



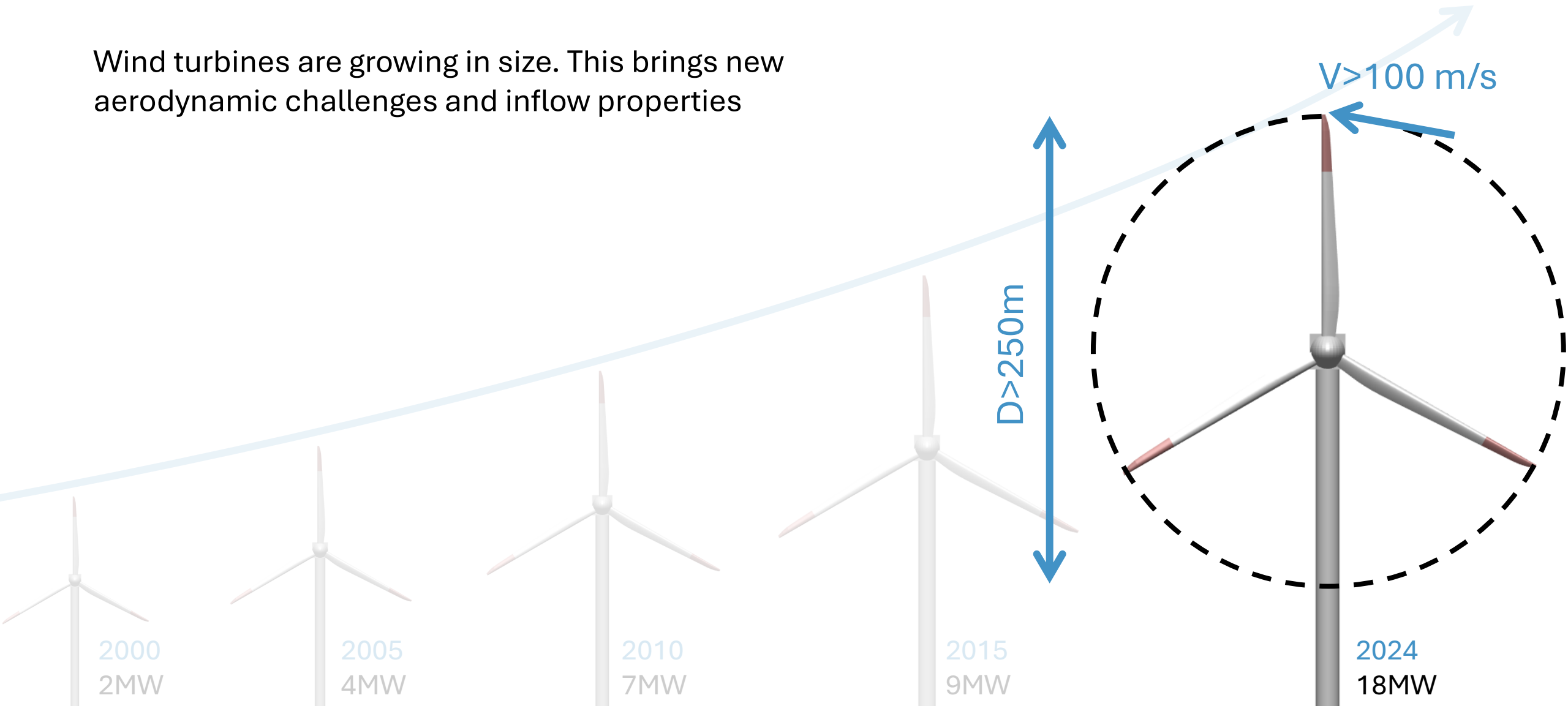
ANY QUESTIONS ON
AIRFOIL REQUIREMENTS



FLOW PROPERTIES

LARGE WIND TURBINES

Wind turbines are growing in size. This brings new aerodynamic challenges and inflow properties



NON-DIMENSIONAL FLOW PROPERTIES

Non-dimensional flow properties are dimensionless numbers that characterize fluid flow by representing the ratio of different physical forces:

- Reynolds number
- Mach number

REYNOLDS NUMBER

Ratio between inertial and viscous forces

$$Re = \frac{\rho UL}{\mu}$$

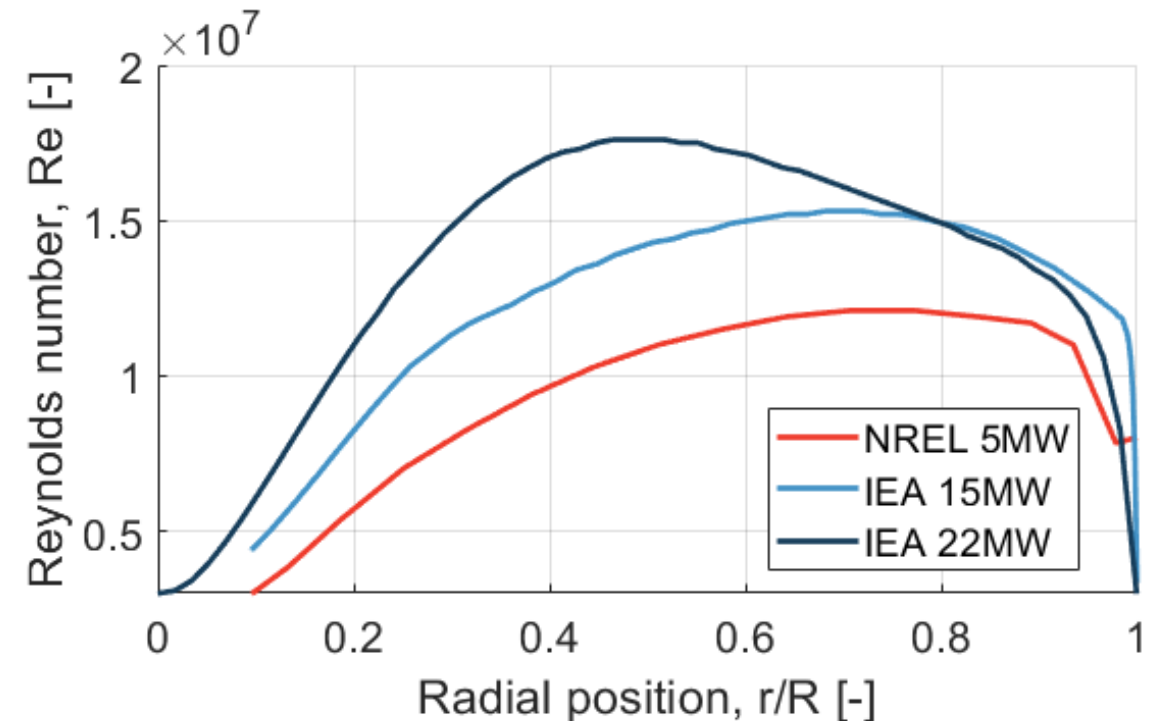
ρ is the fluid density [kg/m³]

U is the flow speed [m/s]

L is a characteristic length [m]

μ is the dynamic viscosity of the fluid [kg/(ms)]

Higher Reynolds number, the lesser viscosity plays a role

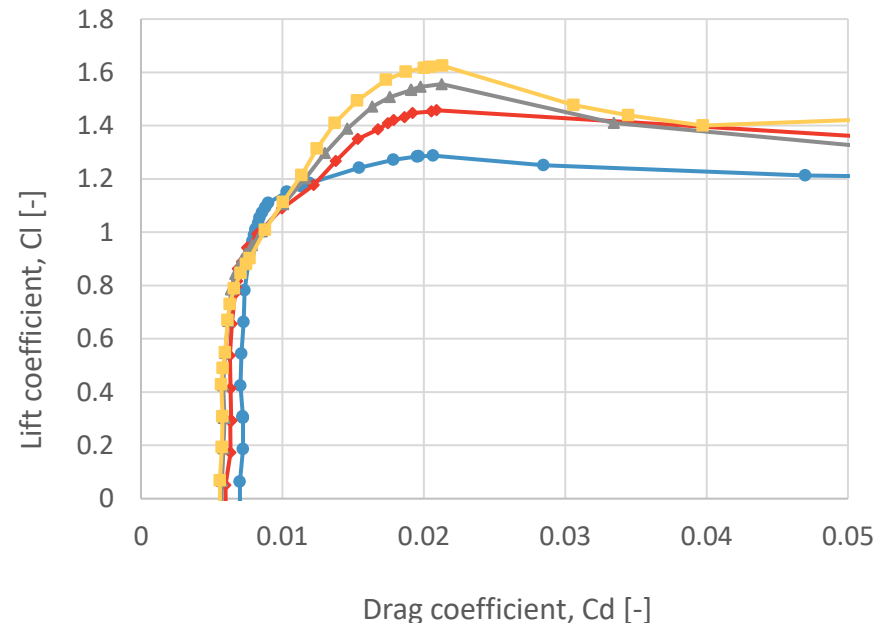
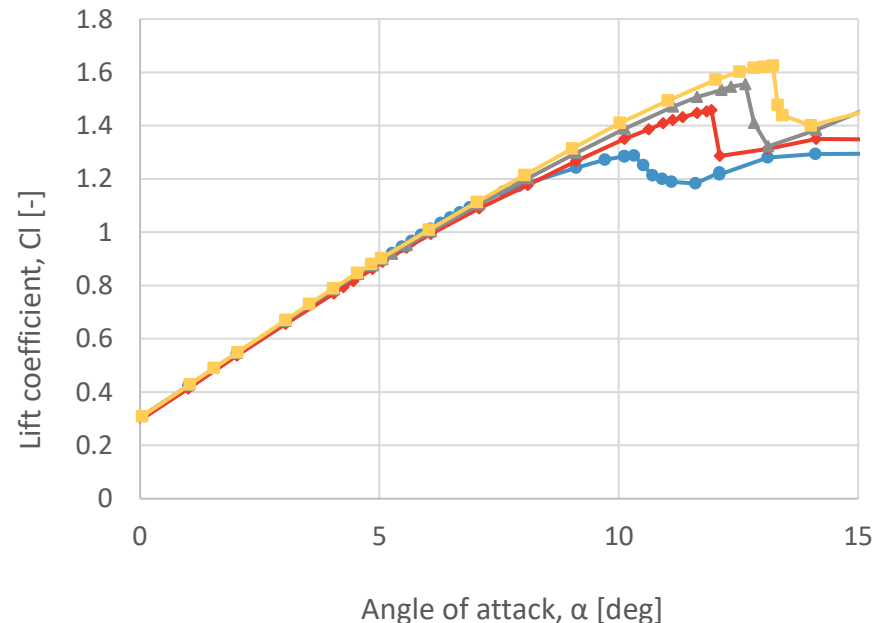


REYNOLDS NUMBER

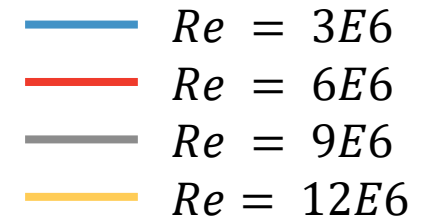
Many effects play when **increasing the Reynolds number**. In general, one may expect:

- Transition point moves forwards as Re has destabilizing effect on boundary layer flow
- Thinner boundary-layer

Higher $C_{l,max}$, higher lift slope $C_{l,\alpha}$, less drag C_d but smaller low-drag range
(C_l/C_d)_{max} will increase, but $C_{l,design}$ will decrease



**Experimental
data DU00W212**



MACH NUMBER

Ratio between flow speed and speed of sound

$$M = \frac{U}{a}$$

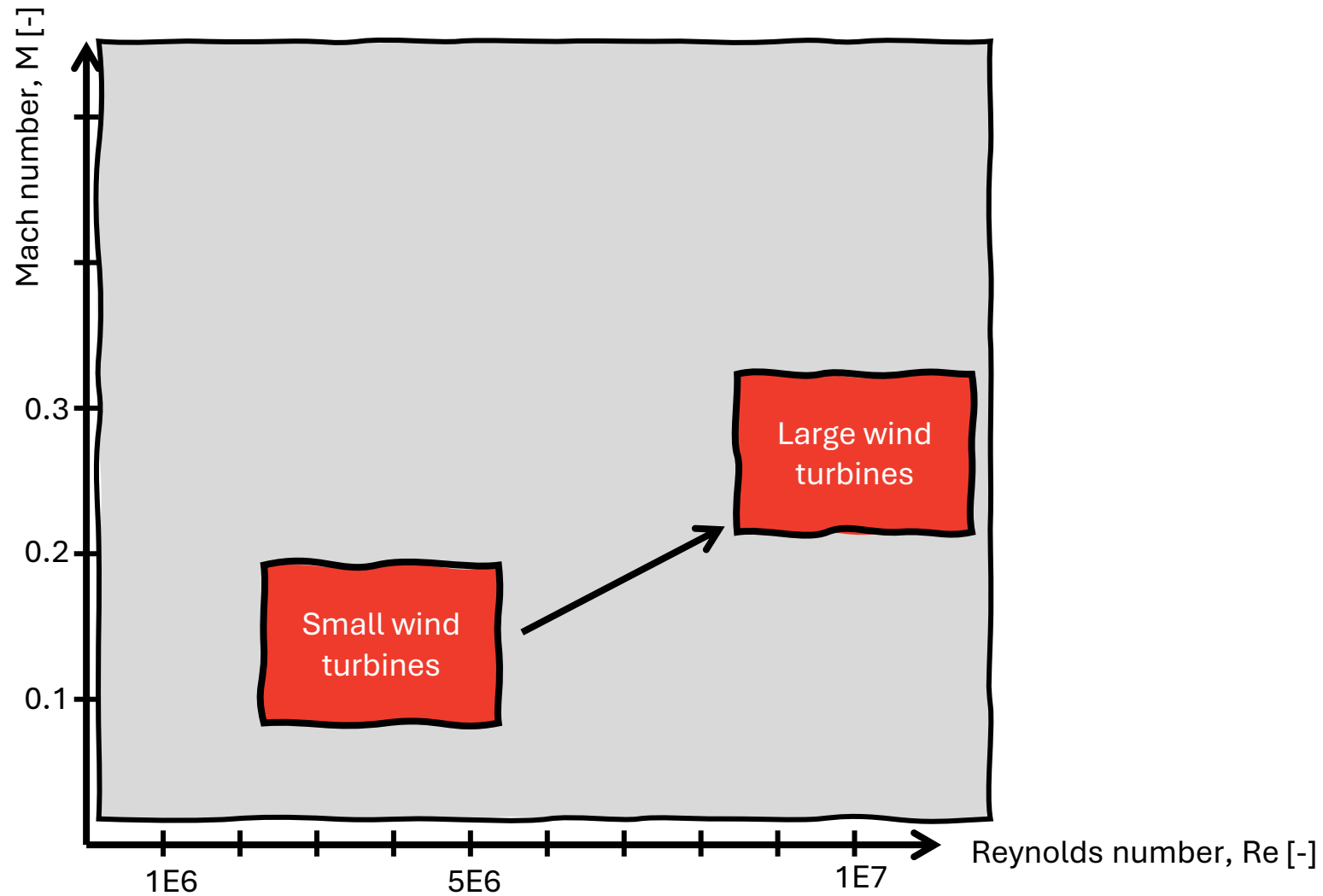
U is the flow speed [m/s]

a is the speed of sound [m/s]

High Mach numbers means that the flow becomes compressible.

Generally, results in a lift reduction

NON-DIMENSIONAL FLOW PROPERTIES



BREAK



ROUGHNESS

ROUGHNESS - SOILING



ROUGHNESS - SOILING



ROUGHNESS - ICING



ROUGHNESS - EROSION



ROUGHNESS - EROSION

1 year in service



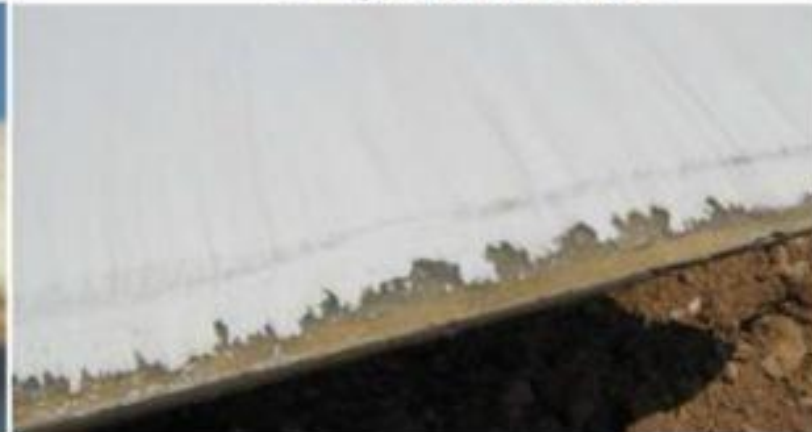
2 years in service



10 years in service



10+ years in service



ROUGHNESS - EROSION

Erosion Level	1	2	3	4
Example Picture				

ROUGHNESS – LEADING EDGE PROTECTION



ROUGHNESS – MANUFACTURING

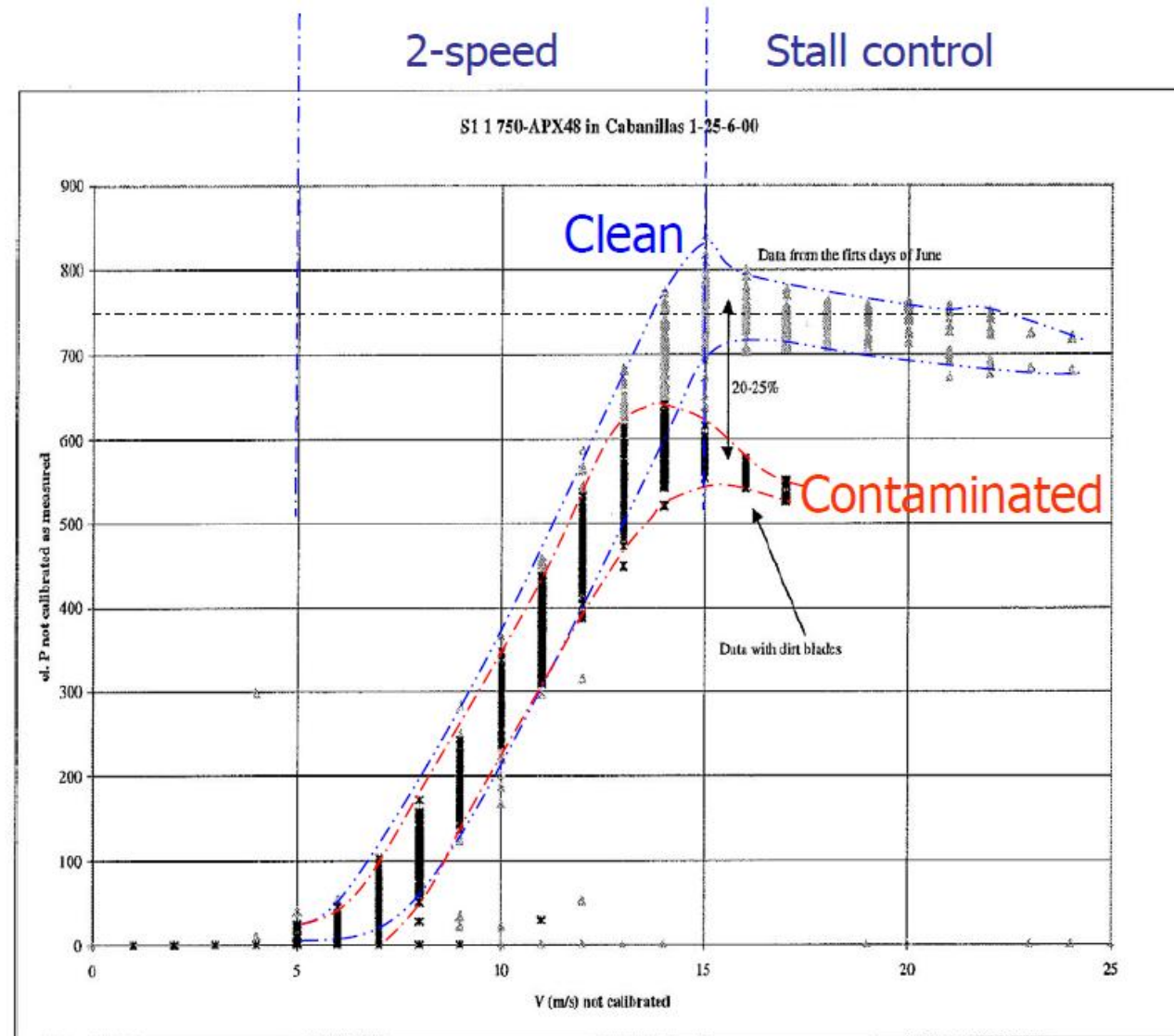


ROUGHNESS – ON THEIR WAY TO REPAIR



TURBINE PERFORMANCE LOSS

Stall-controlled
turbine

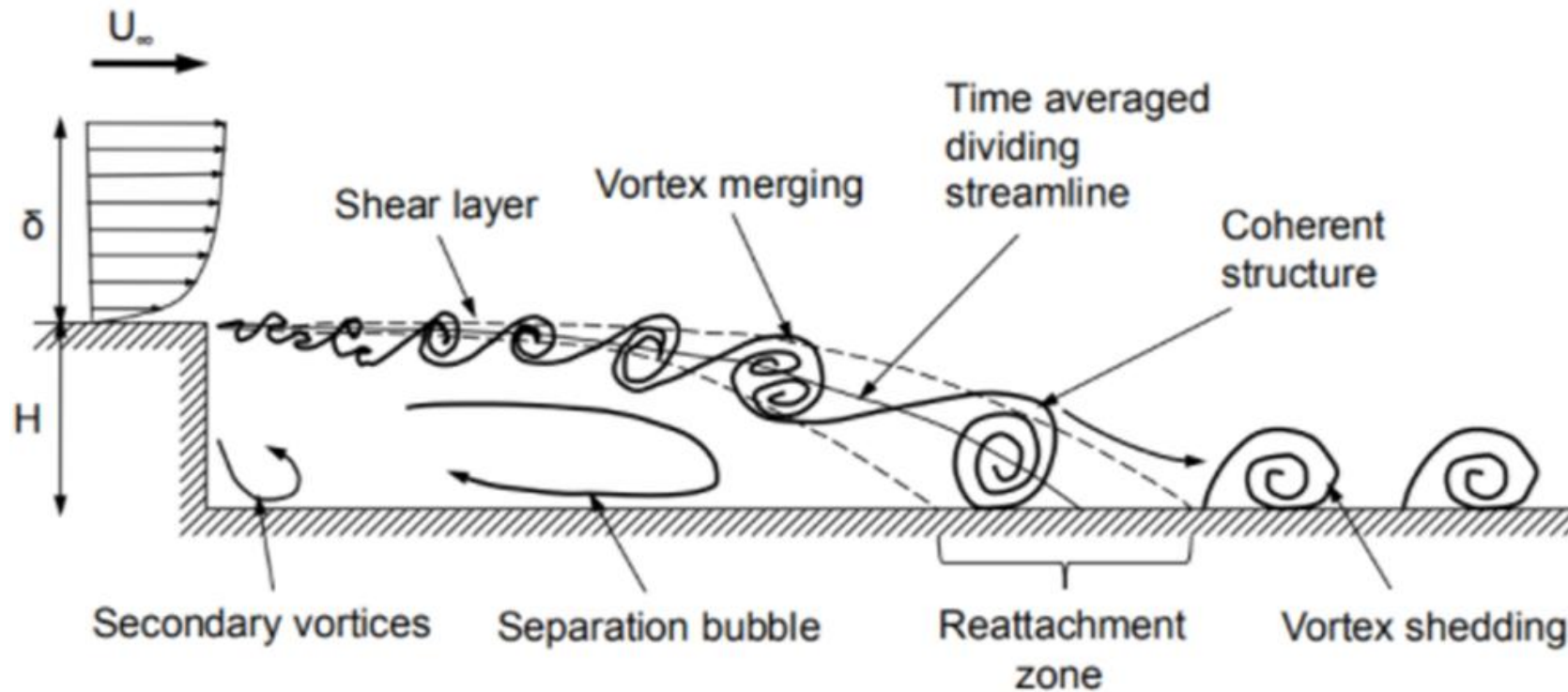


TURBINE PERFORMANCE LOSS

Turbine	Annual Yield [GWh]	10% energy loss [GWh]	Financial loss at 0.26EUR/kWh [euro]
V90	8.69	0.869	225 940
S88	9.48	0.948	246 480
NREL 5MW	19.61	1.961	509 860

AIRFOIL ROUGHNESS EFFECT

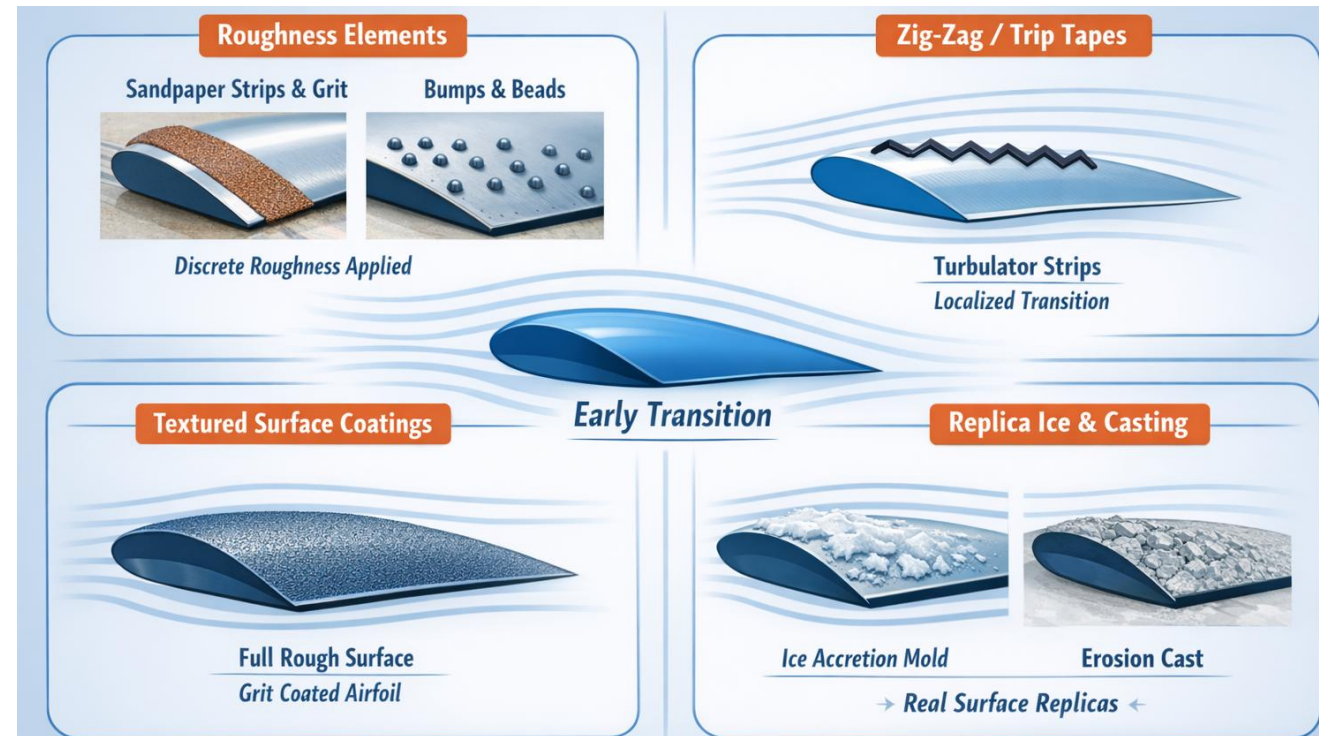
Roughness generally triggers the flow to **transition early**. This means that the transition point will move forward and the **boundary layer thickens**. The airfoil will encounter **more drag**, a **reduction in lift** and **stall occurs faster**.



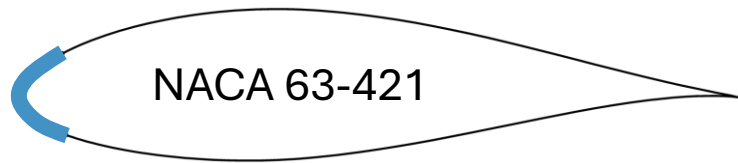
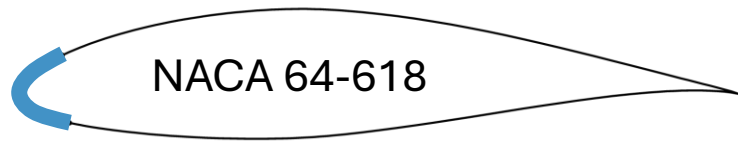
ROUGHNESS REPLICATION

Many ways are used to **replicate roughness** in experimental testing:

- Real surface replica
- ZZ-tape: at various chordwise locations
- Distributed bumps
- Sandpaper-like structure

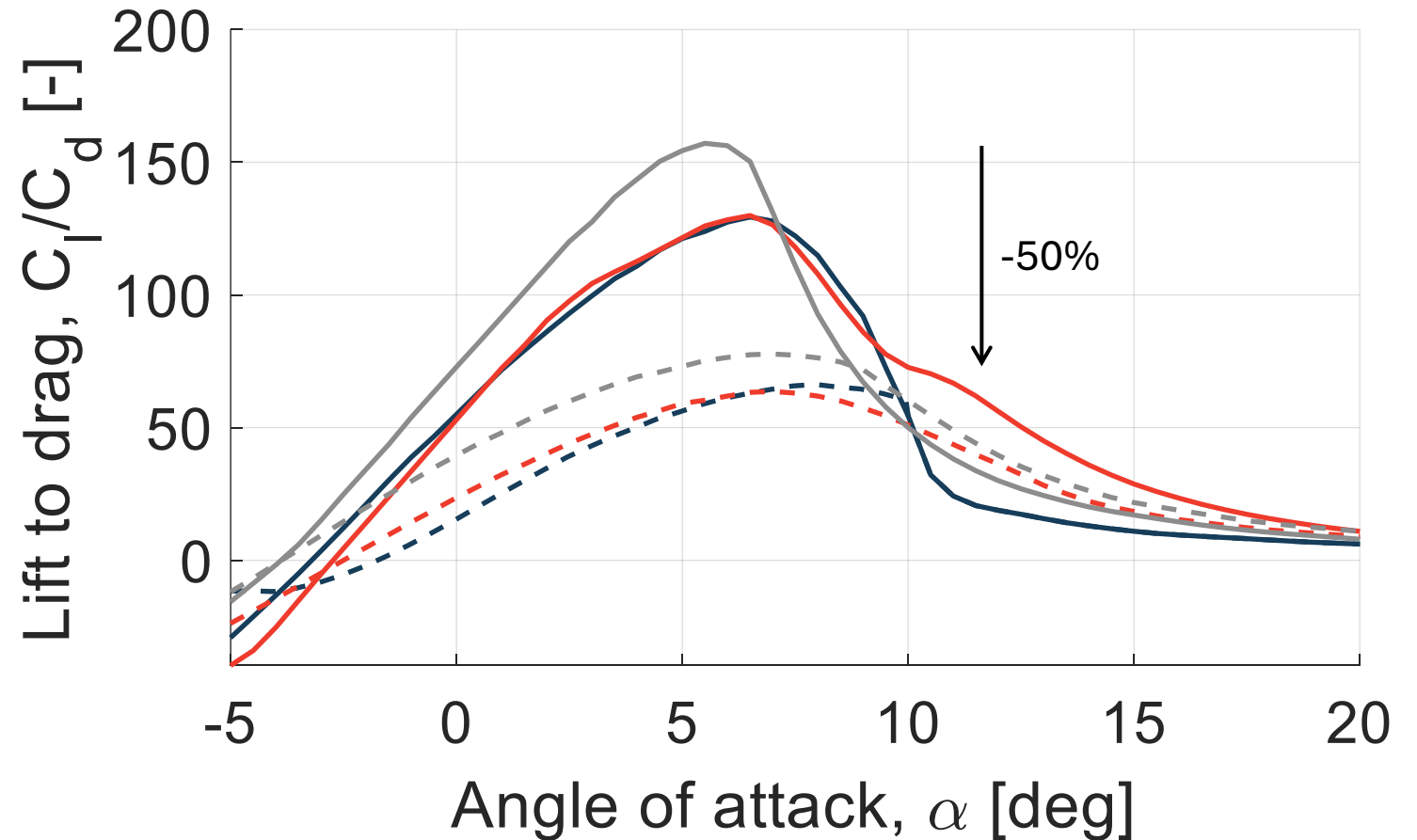
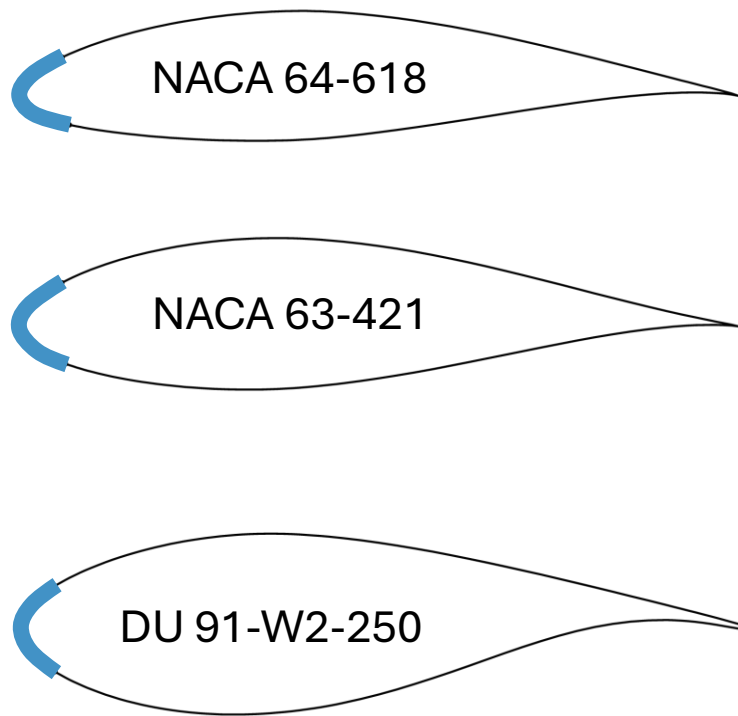


AIRFOIL PERFORMANCE LOSS

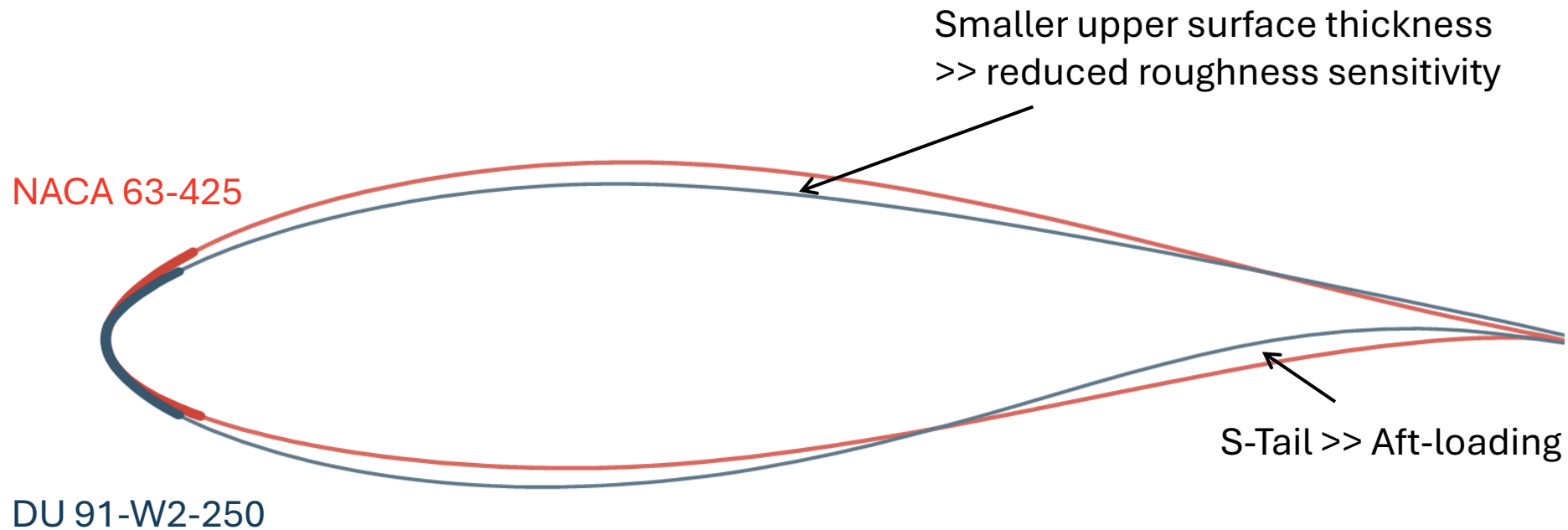


$$C_{p,\max} = \left(\frac{16}{27}\right) \lambda \left[\lambda + \frac{1.32 + \left(\frac{\lambda-8}{20}\right)^2}{B^{\frac{2}{3}}} \right]^{-1} - \frac{(0.57)\lambda^2}{\frac{C_l}{C_d} \left(\lambda + \frac{1}{2B} \right)}$$

AIRFOIL PERFORMANCE LOSS



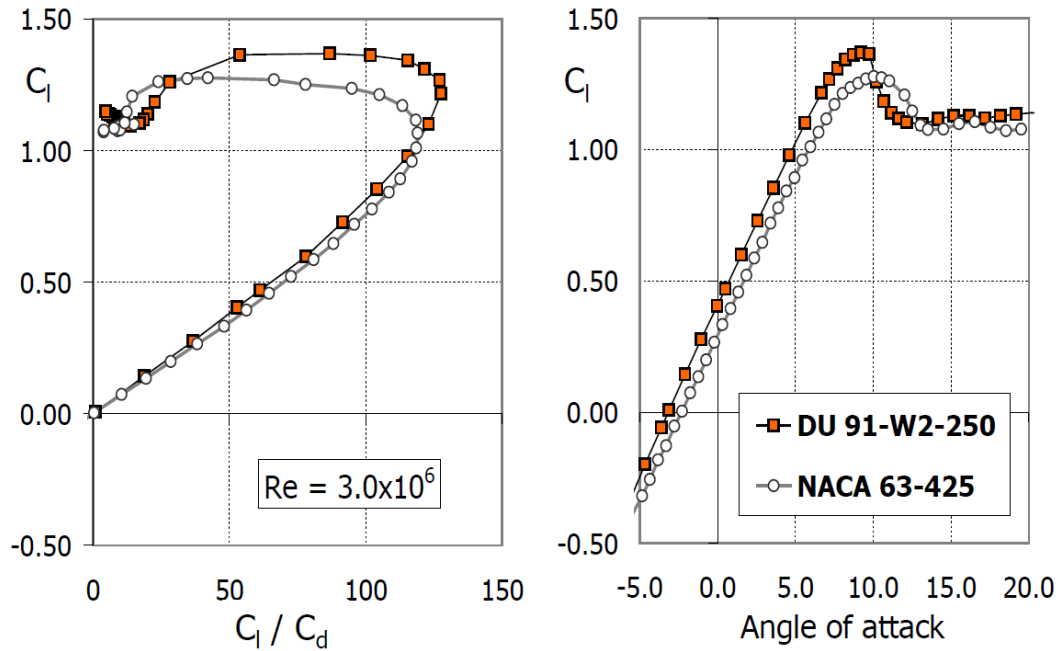
AIRFOIL PERFORMANCE LOSS



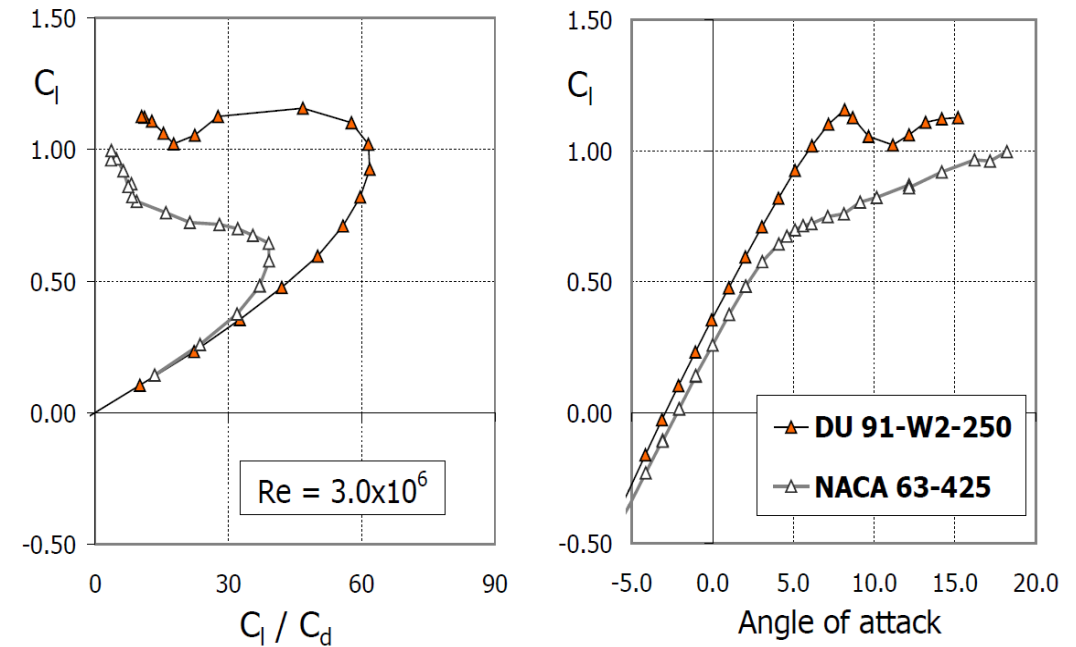
Modern wind turbines (such as the DU91-W2-250) are designed considering they should be **roughness insensitive**.

AIRFOIL PERFORMANCE LOSS

Clean performance



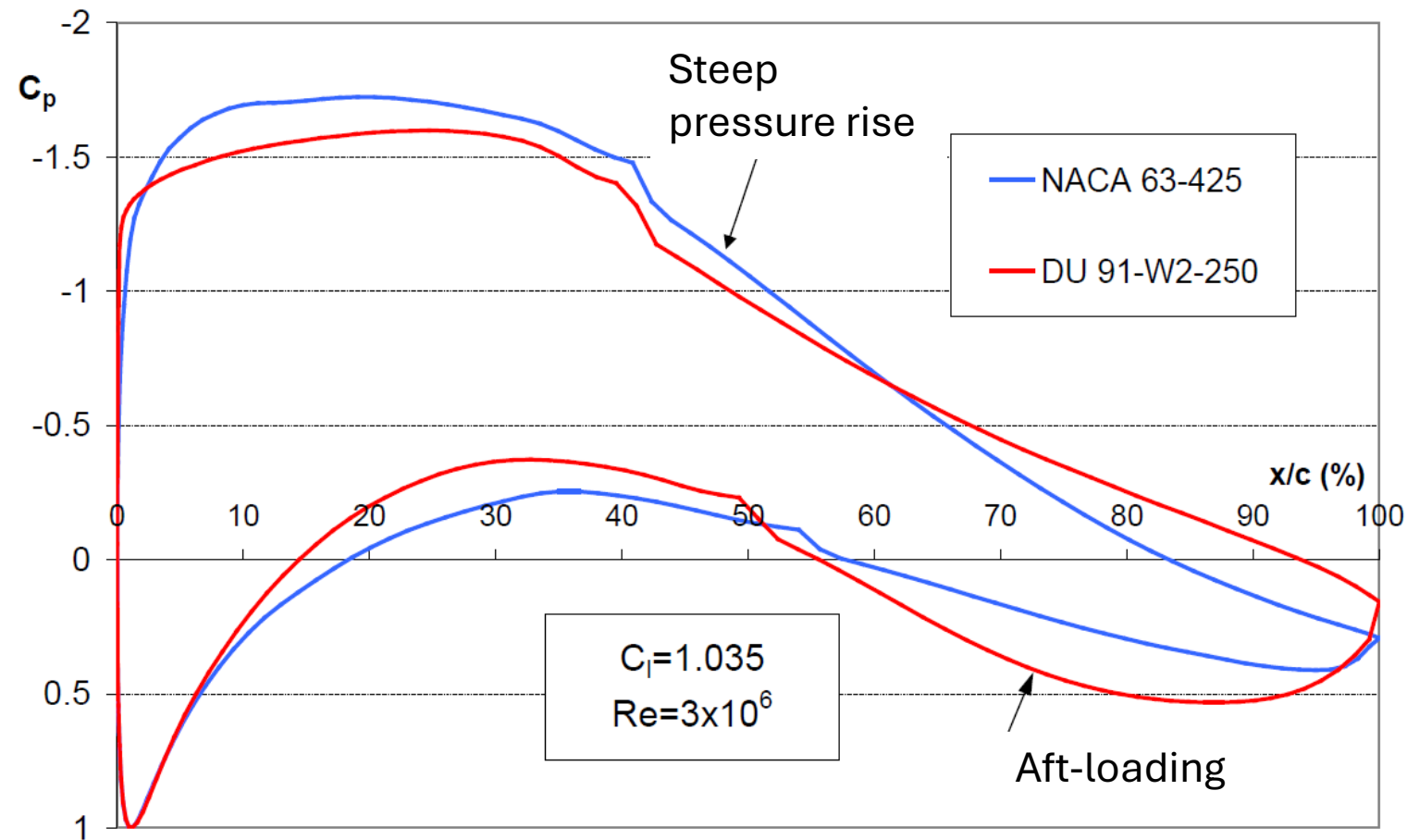
Rough performance (using ZZ-tape at 5%)



Modern wind turbines (such as the DU91-W2-250) are designed considering they should be **roughness insensitive**.

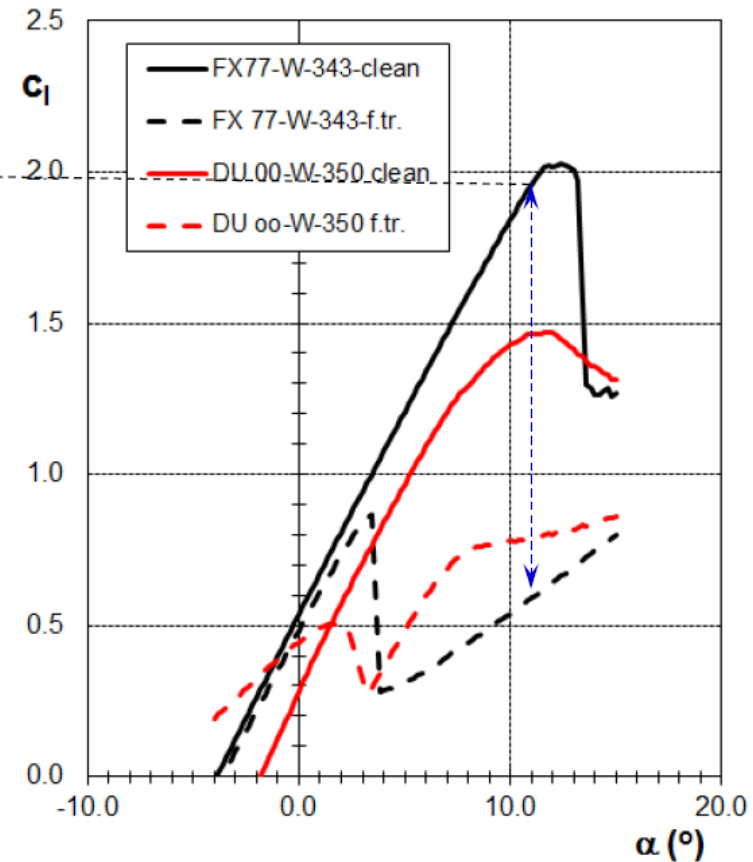
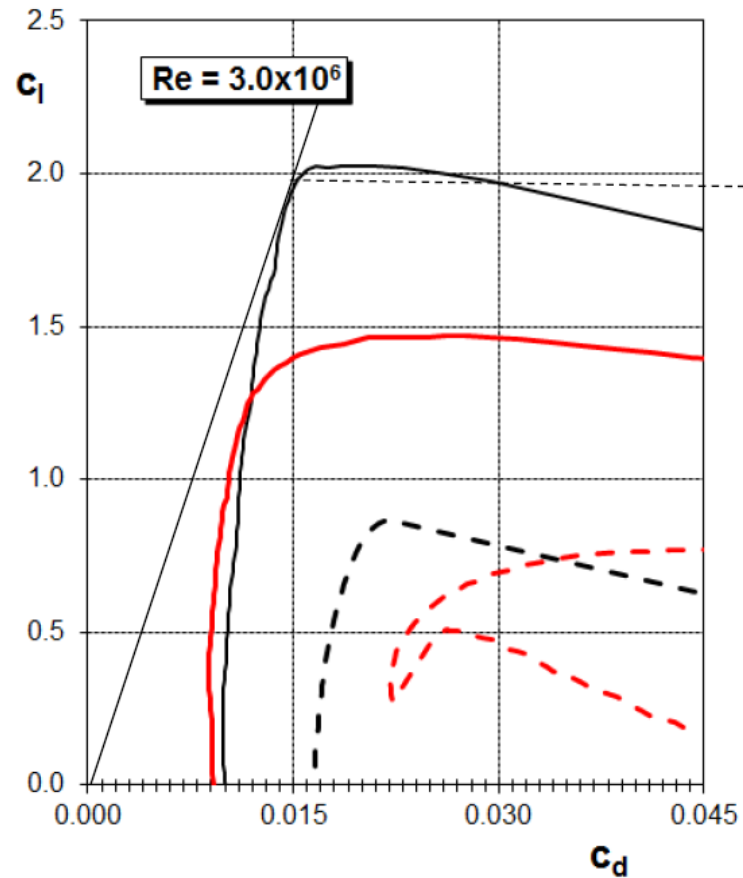
AIRFOIL PERFORMANCE LOSS

Pressure distribution for the same C_l -value



AIRFOIL PERFORMANCE LOSS

Thick airfoils see a significant reduction in performance: decrease in $C_{l,max}$ and increase in C_d in case of roughness

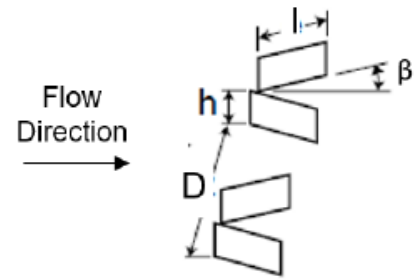


VORTEX GENERATORS

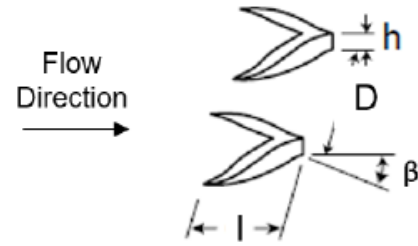


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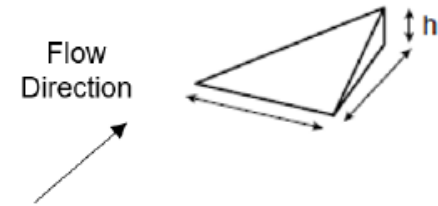
VORTEX GENERATORS



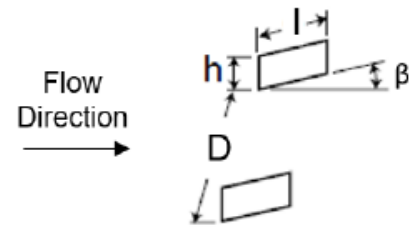
Counter-rotating Vanes



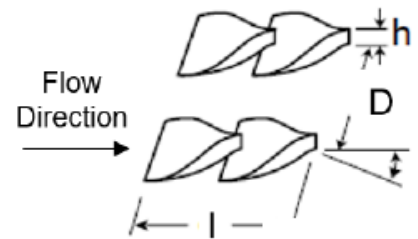
Wishbone



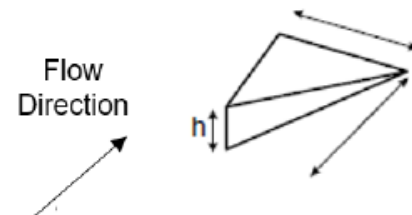
Forward Wedge



Co-rotating Vanes



Doublet



Backward Wedge

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VORTEX GENERATORS

Without VGs



Zero angle of attack



Medium angle of attack



Stall angle of attack

With VGs



Zero angle of attack



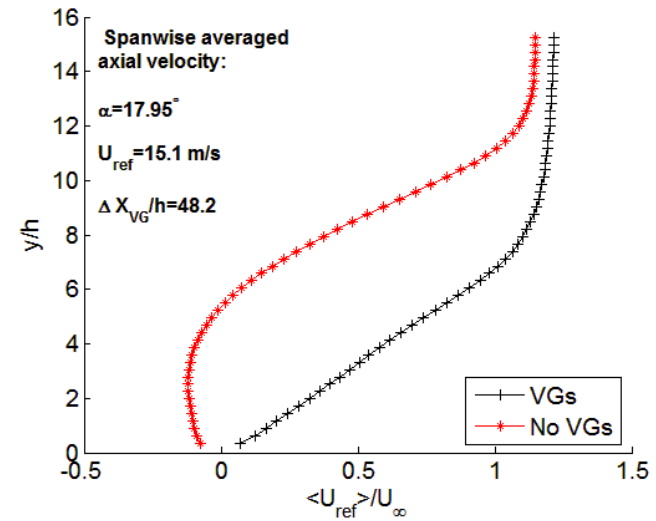
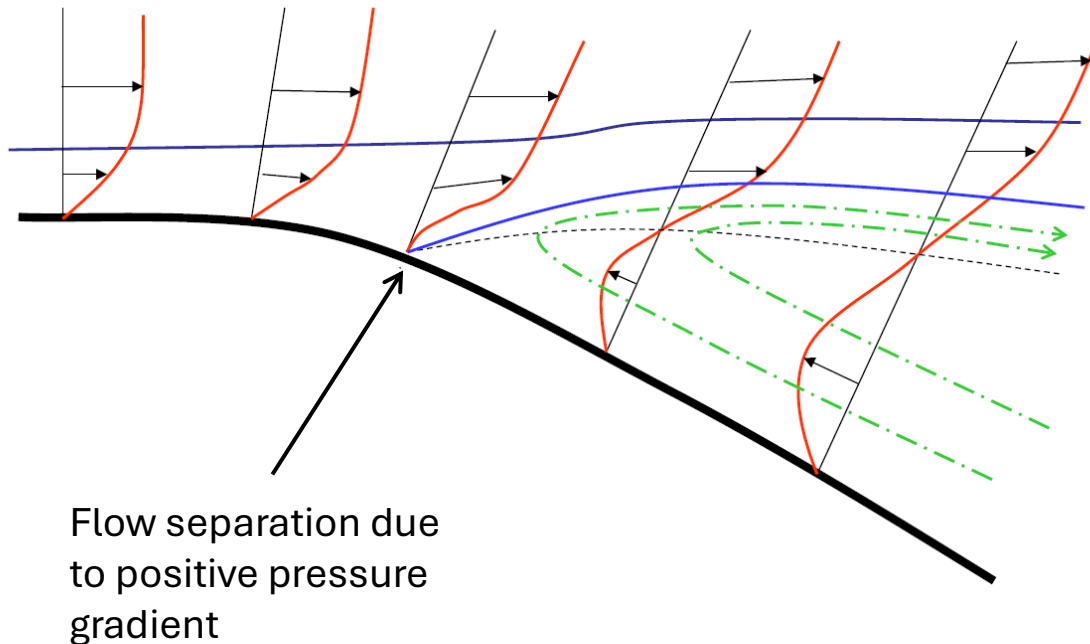
Medium angle of attack



Stall angle of attack

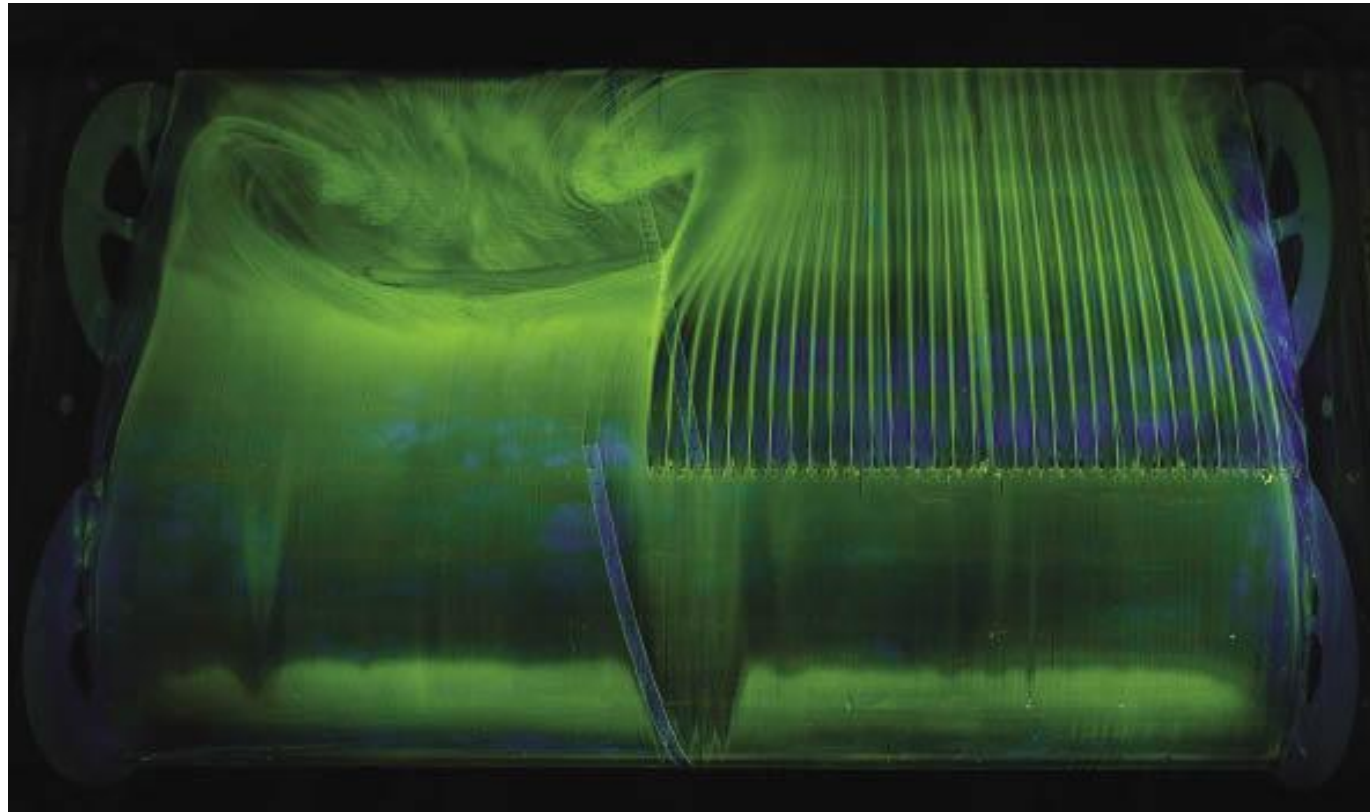
VORTEX GENERATORS

Vortex generators create **streamwise vortices** from the free tips of the VGs. These vortices enhance mixing between the high-energy flow in the outer part of the boundary layer with the low-energy regions near the walls, and re-energize the flow close to the surface. This makes the boundary layer more resistant against flow separation.



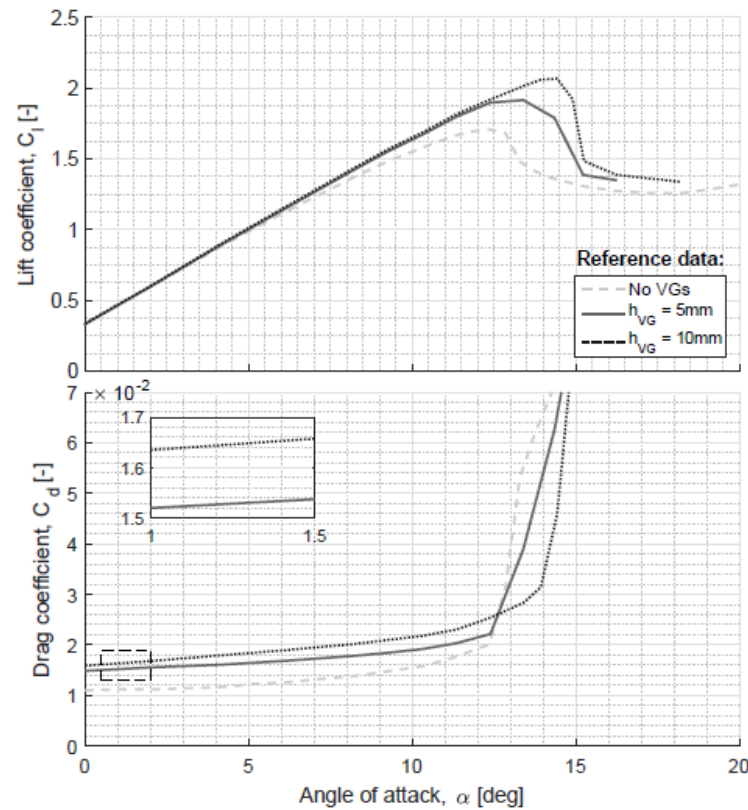
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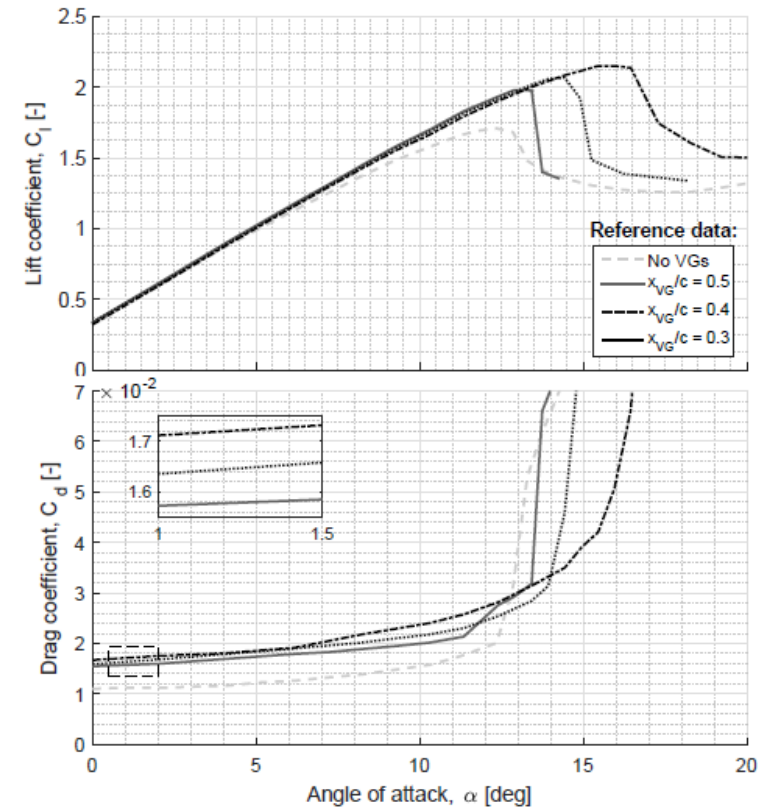


VORTEX GENERATORS

VGs can enhance the maximum lift, but at a cost of drag in the parts where VGs are not yet useful

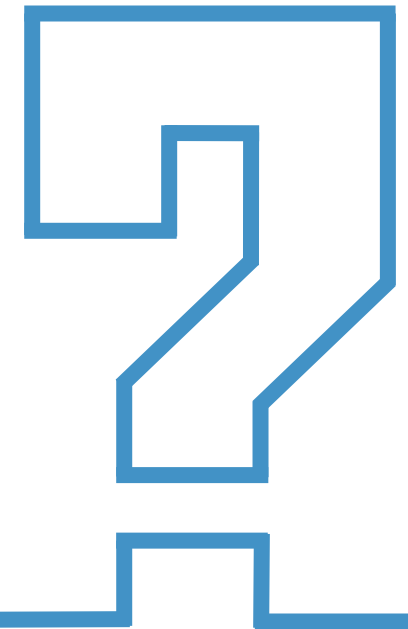


(a) Reference data



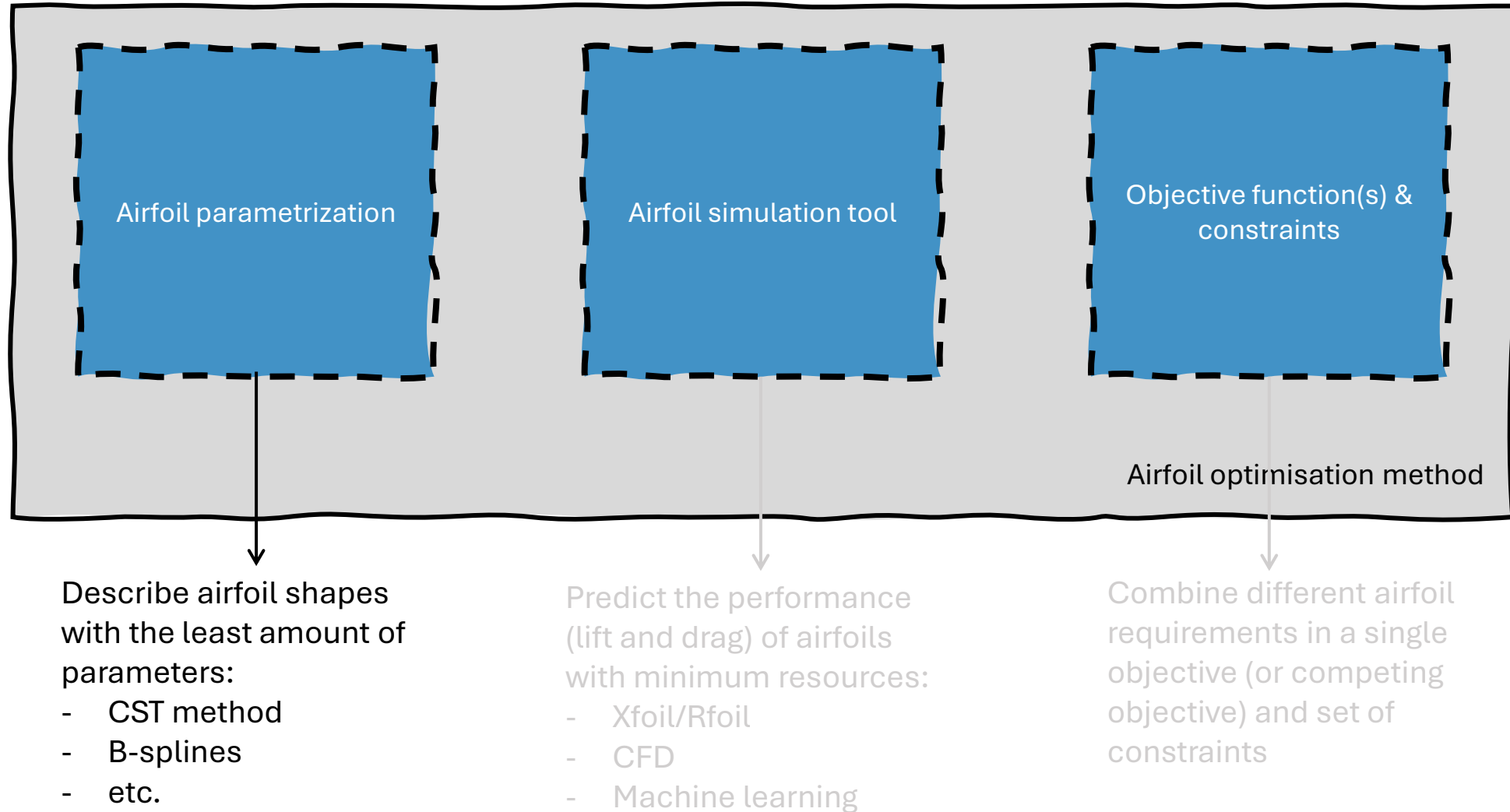
(a) Reference data

ANY QUESTIONS ON
ROUGHNESS & VGs

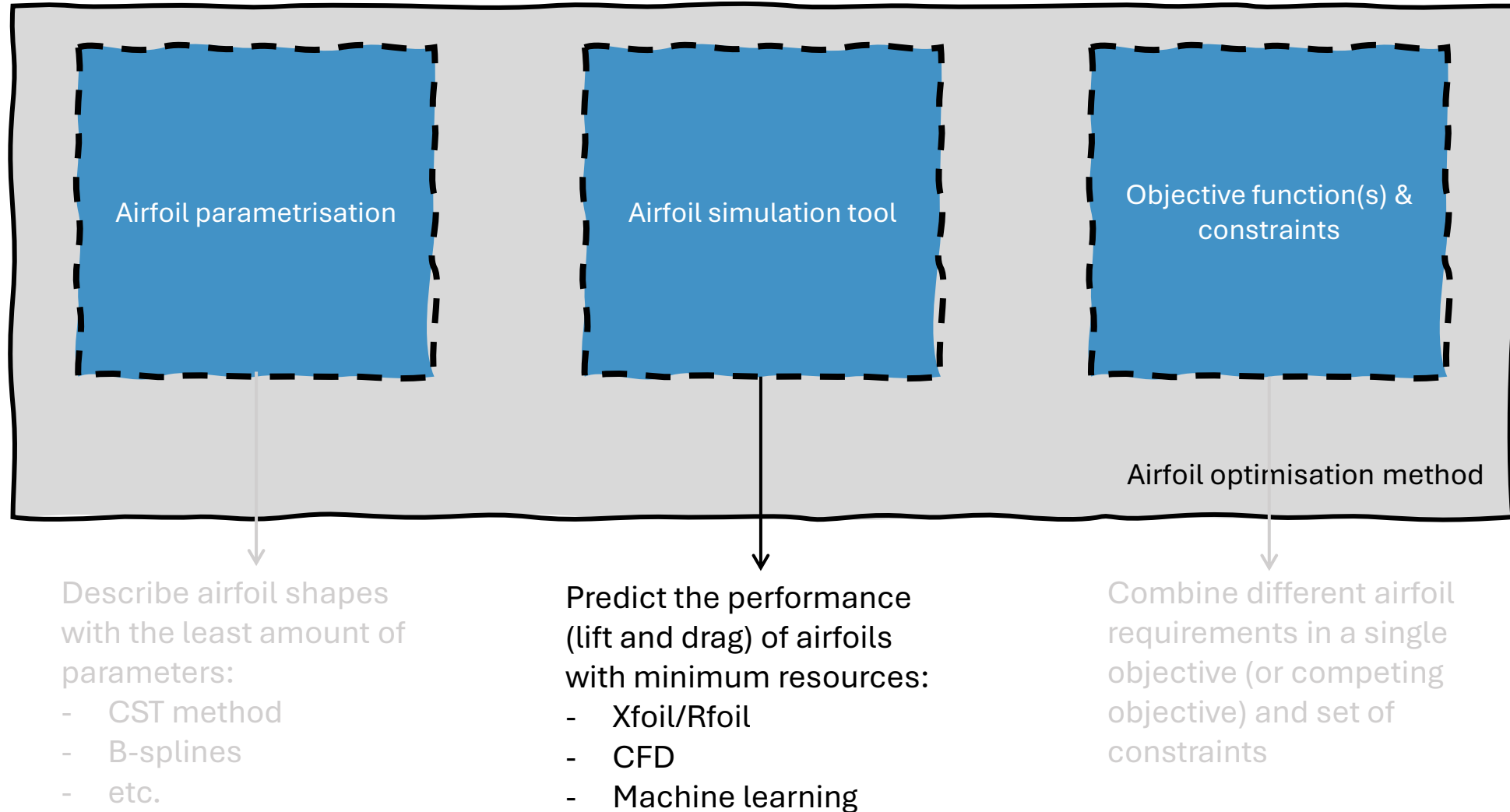


AIRFOIL DESIGN AND EVALUATION

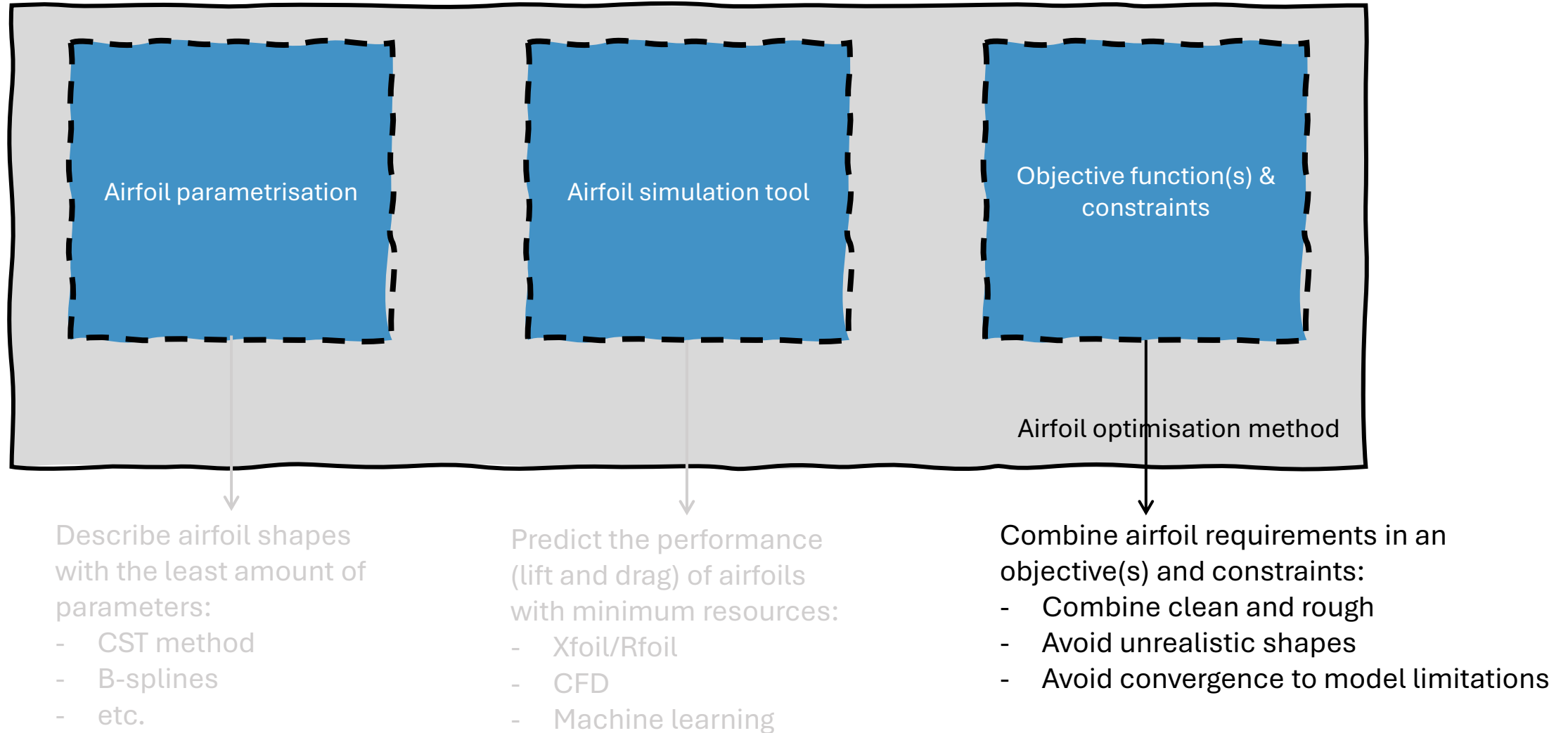
AIRFOIL DESIGN AND OPTIMISATION



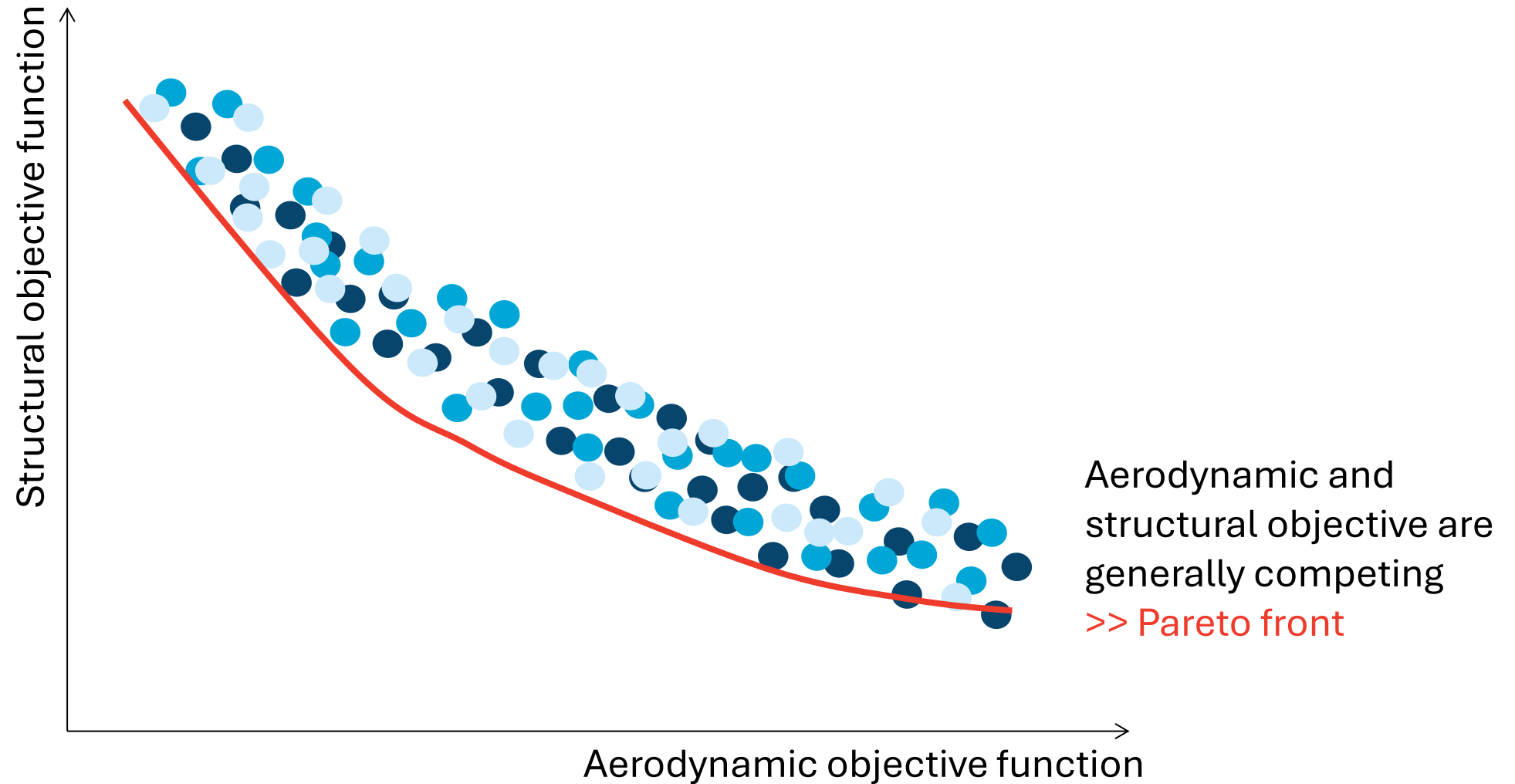
AIRFOIL DESIGN AND OPTIMISATION



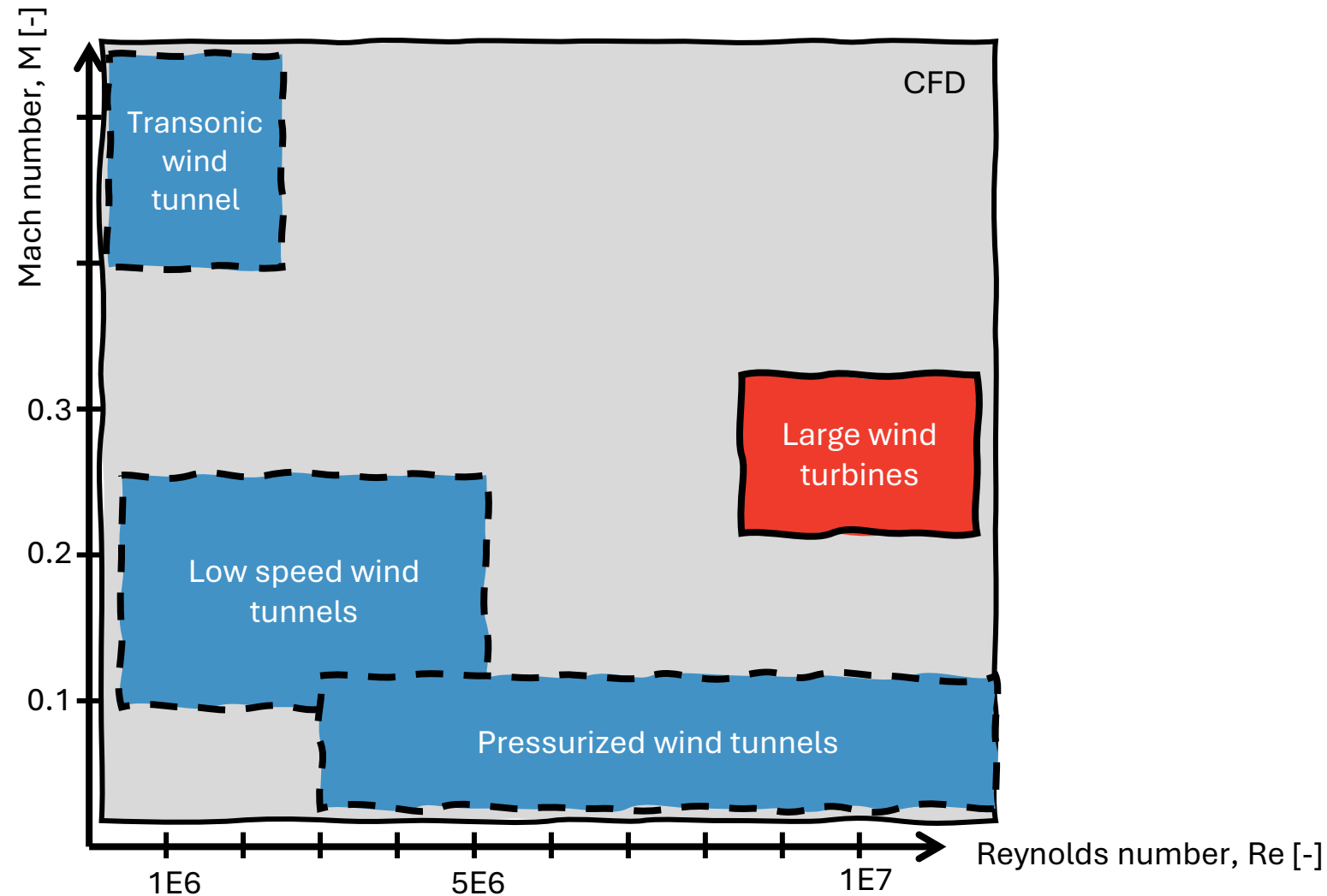
AIRFOIL DESIGN AND OPTIMISATION



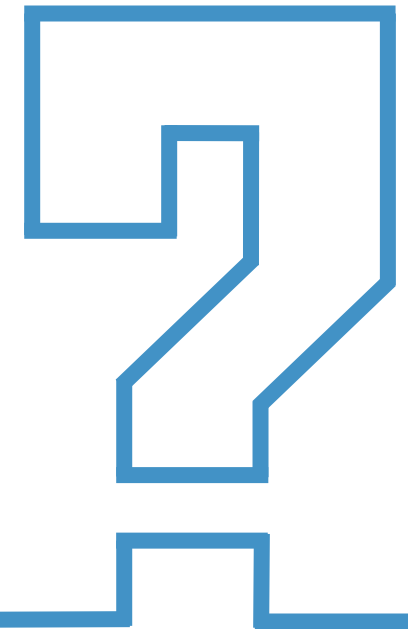
AIRFOIL DESIGN AND OPTIMISATION



AIRFOIL EVALUATION – WIND TUNNEL TESTING

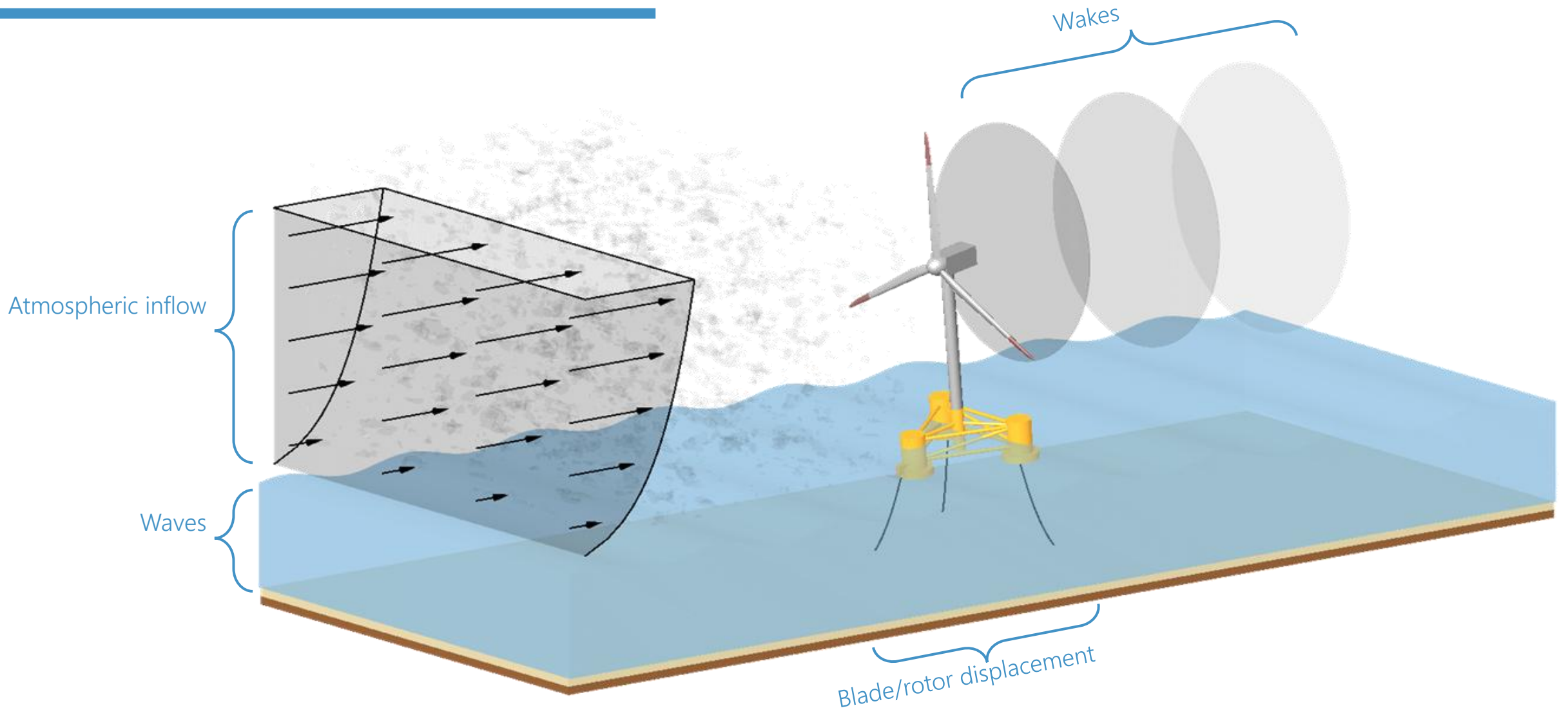


ANY QUESTIONS ON
AIRFOIL OPTIMISATION



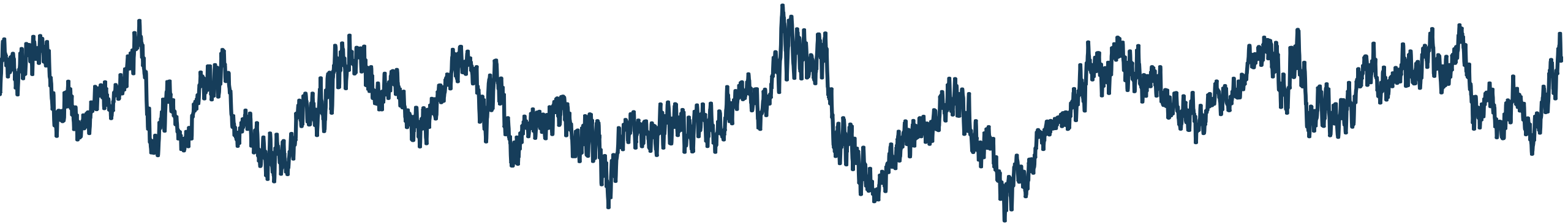
STEADY VS UNSTEADY

TURBULENT INFLOW

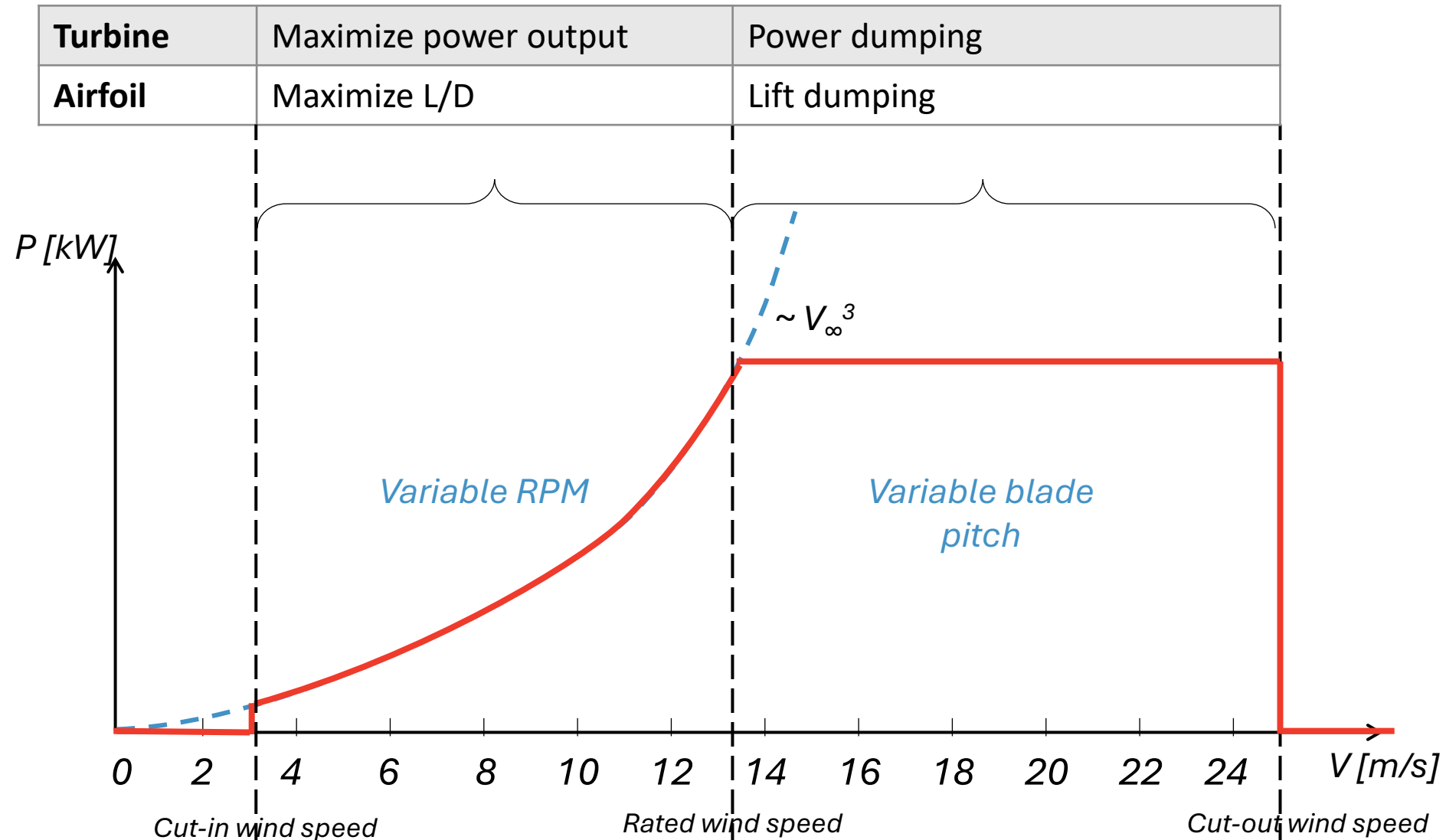


TURBULENT INFLOW

There are large variation in relative wind speed and inflow angle at high frequencies, therefore steady lift and drag polars only provide a part of the story.

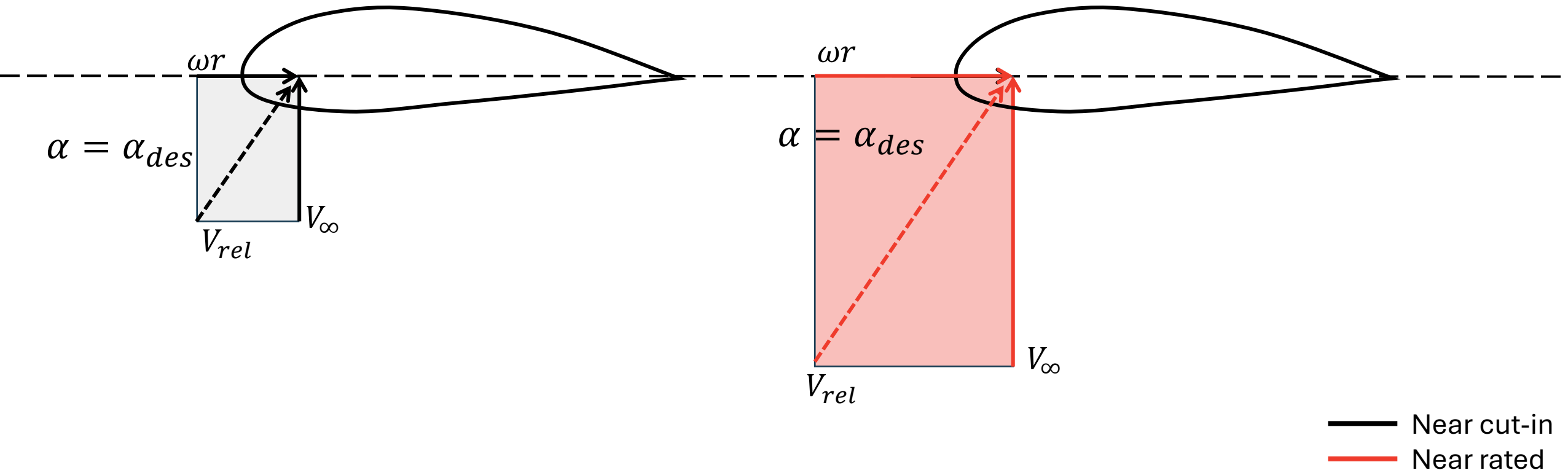


TURBINE OPERATION



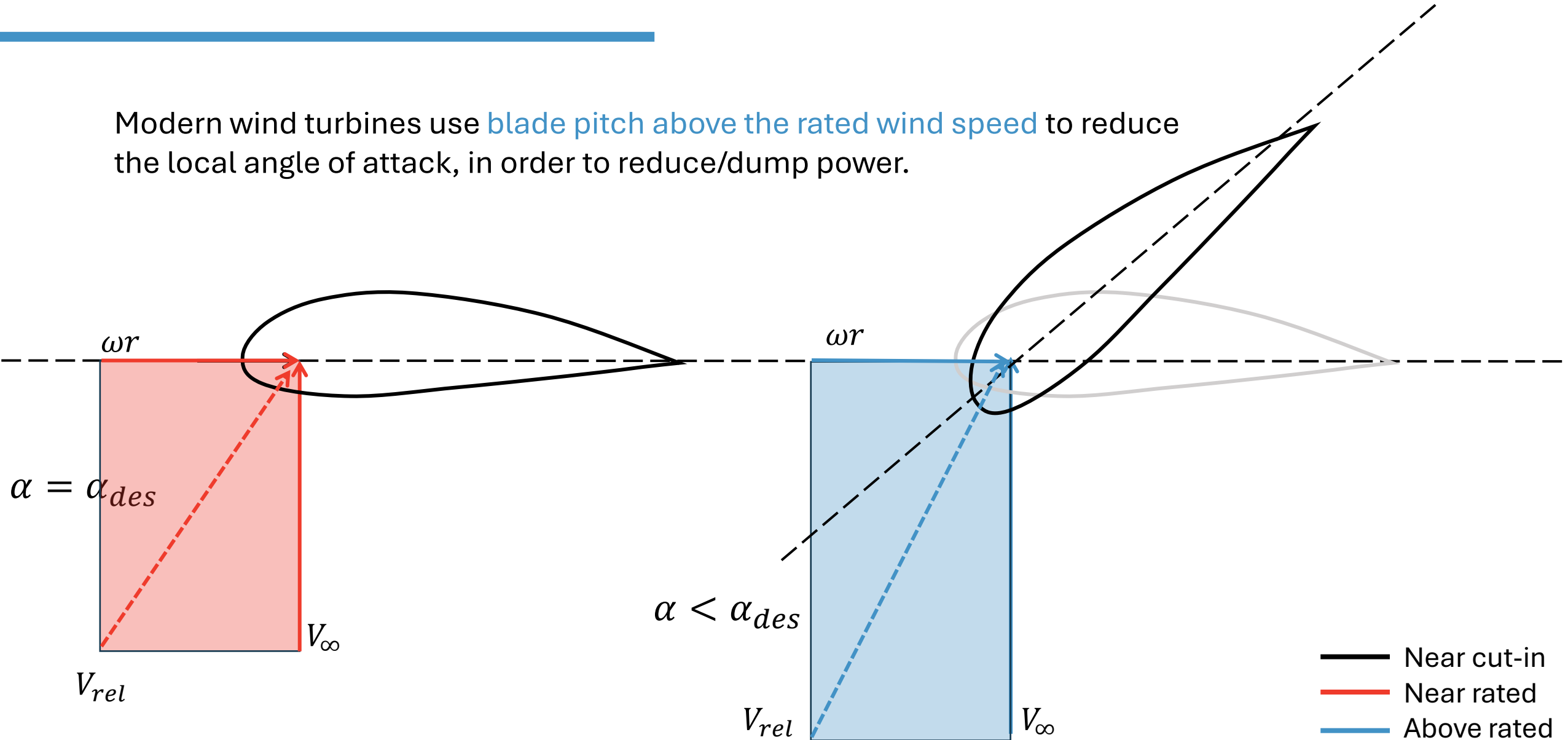
TURBINE OPERATION – BELOW RATED

Modern wind turbines use **variable RPM below the rated wind speed** to keep the local angle of attack constant and equal to the design angle of attack, in order to maximize power output.



TURBINE OPERATION – ABOVE RATED

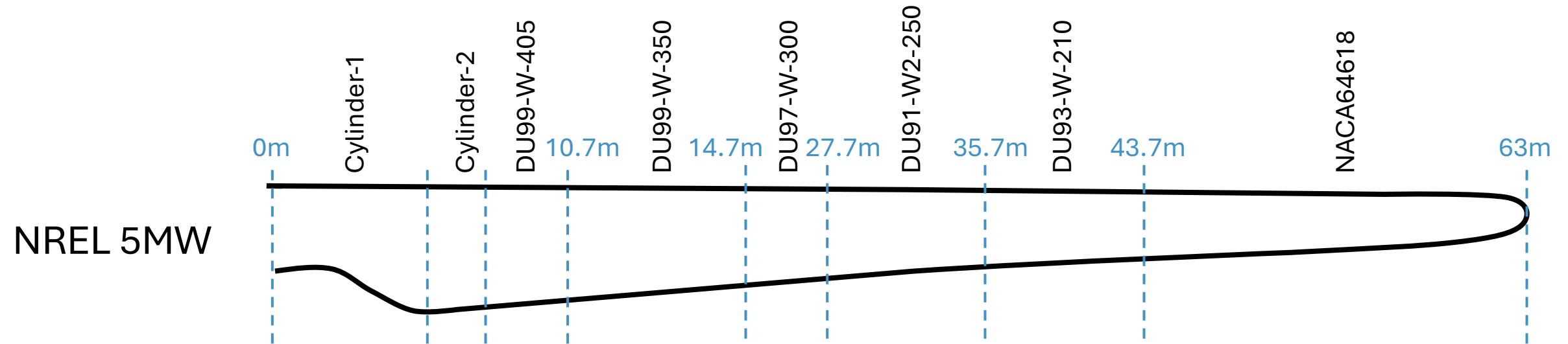
Modern wind turbines use **blade pitch above the rated wind speed** to reduce the local angle of attack, in order to reduce/dump power.



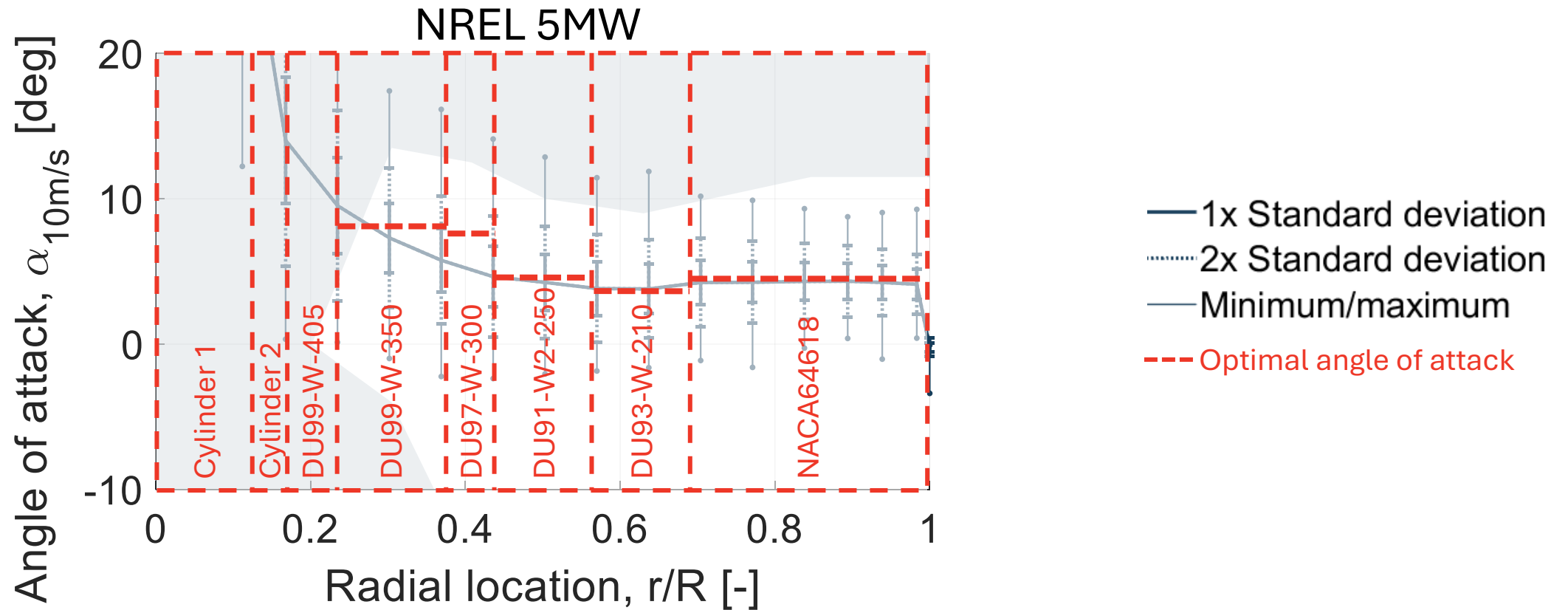
TURBINE OPERATION – NREL 5MW

“Real” operating conditions of the NREL 5MW reference turbine:

- Turbulence
- Shear
- Aero-elastic deflection
- Control
- Etc.

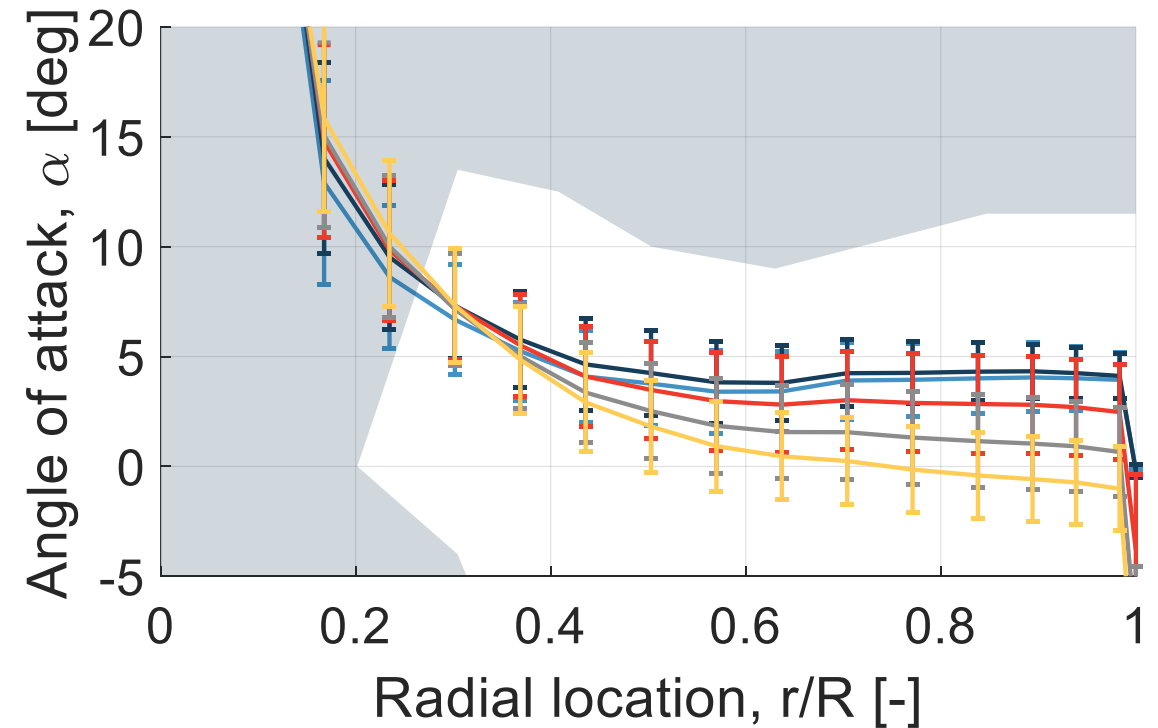
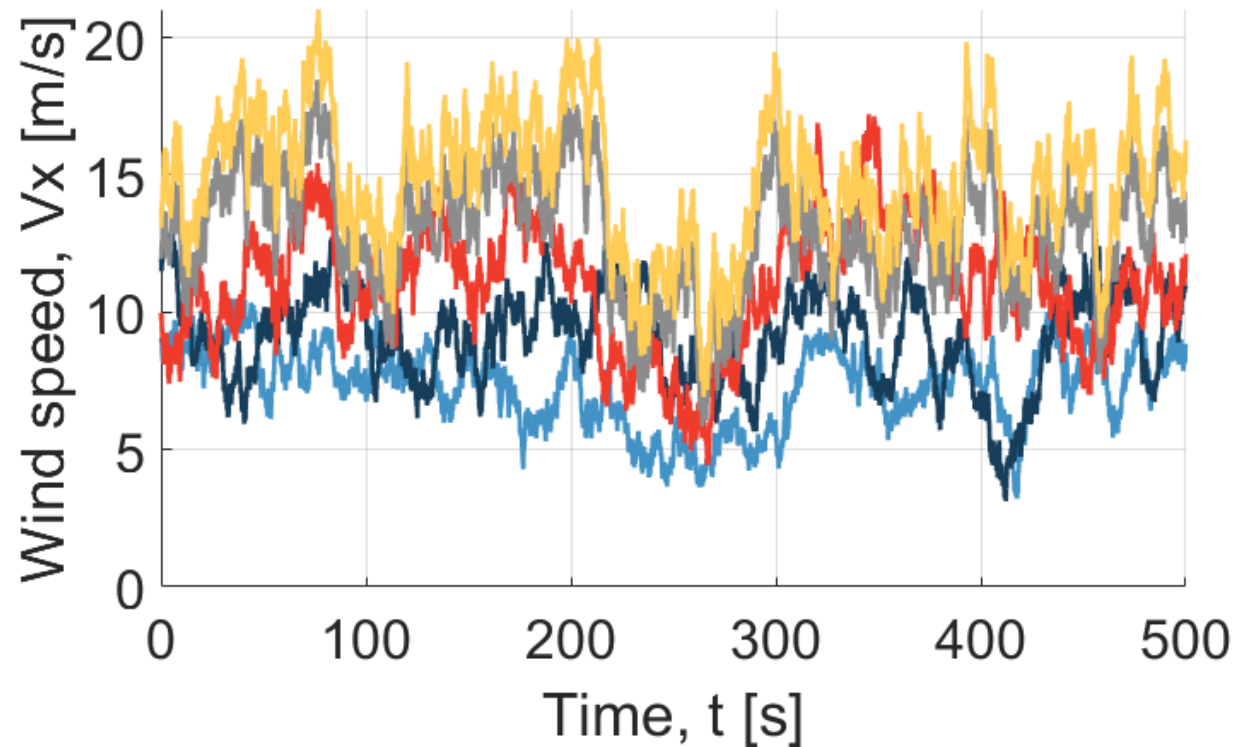


TURBINE OPERATION – NREL 5MW



TURBINE OPERATION – NREL 5MW

- $V_{\infty} = 8 \text{ m/s}$
- $V_{\infty} = 10 \text{ m/s}$
- $V_{\infty} = 12 \text{ m/s}$
- $V_{\infty} = 14 \text{ m/s}$
- $V_{\infty} = 16 \text{ m/s}$

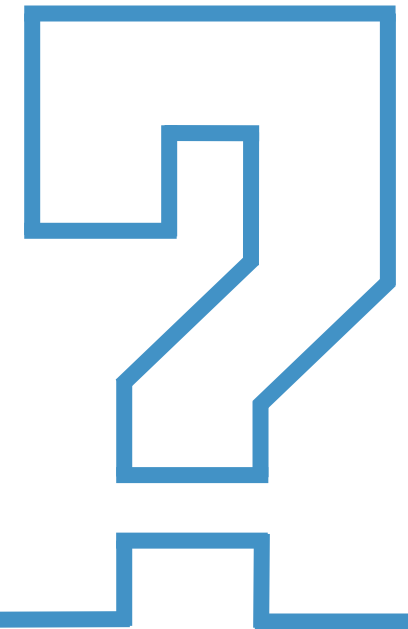


STEADY VS UNSTEADY

Wind turbines are inherently unsteady:

- Designing/optimizing for a single point might not be ideal as wind turbines see a [range of angles of attack](#)
- Below and above rated, the angle of attacks are significantly different
- Steady aerodynamics is only one side of the story, [unsteady airfoil aerodynamics](#) should be considered, e.g.
 - Hysteresis
 - Dynamic stall
- 3D effects, centrifugal and coriolis effects, etc.

ANY QUESTIONS ON
TURBINE OPERATION



SUMMARY

LEARNING OBJECTIVES

At the end of this lecture, you should be able to:

- Describe the 2D velocity and force triangle
- Derive the desired airfoil characteristics along the blade
- Understand the role of inflow properties on airfoil behaviour
- Understand the role of roughness on airfoil performance
- Describe the approach for airfoil design and evaluation

Keywords: (1) Velocity and force triangle, (2) Airfoil requirements, (3) Flow properties, (4) Roughness, (5) Airfoil design and evaluation